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CHAPTER 4

REACTION KINETICS

SUGGESTED TIME: 12 PERIODS

IN THIS CHAPTER

This chapter introduces the concepts, experimental techniques, mathematical laws, and theoretical models that allow chemists to quantify and predict the rates of chemical reactions. It equips students with the tools to analyse kinetic data, determine reaction mechanisms, and evaluate the role of catalysts.

- Define reaction rate, average rate, instantaneous rate, and initial rate, and state their units.
- Identify and explain the factors that affect reaction rates (concentration, temperature, surface area, catalyst).
- Design and interpret experiments using chemical (titrimetry) and physical (colorimetry, conductimetry, dilatometry, gas syringe, pressure) methods to follow reaction progress.
- Write the rate law for a reaction, define rate constant, partial orders, and overall order, and determine their units.
- Determine reaction order and rate constant using the initial-rates method and graphical methods (including half-life for first-order reactions).
- Apply collision theory, Maxwell-Boltzmann distribution, and the Arrhenius equation to explain temperature dependence of rates.

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METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

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CHAPTER OBJECTIVES

- Define reaction rate, average rate, instantaneous rate, and initial rate, and state their units.
- Identify and explain the factors that affect reaction rates (concentration, temperature, surface area, catalyst).
- Design and interpret experiments using chemical (titrimetry) and physical (colorimetry, conductimetry, dilatometry, gas syringe, pressure) methods to follow reaction progress.
- Write the rate law for a reaction, define rate constant, partial orders, and overall order, and determine their units.
- Determine reaction order and rate constant using the initial-rates method and graphical methods (including half-life for first-order reactions).
- Apply collision theory, Maxwell-Boltzmann distribution, and the Arrhenius equation to explain temperature dependence of rates.
- Describe transition-state theory, the concept of the activated complex, and compare it with collision theory.
- Distinguish homogeneous and heterogeneous catalysis, explain their mechanisms, and evaluate their industrial relevance.

PREREQUISITES

- Mole concept, stoichiometry, and concentration calculations (Lower Sixth).
- Basic algebra, logarithms, and graph plotting (mathematics).
- Understanding of energy changes in reactions (enthalpy, activation energy) from energetics chapter.
- Familiarity with common laboratory apparatus (burette, spectrophotometer, conductivity meter, gas syringe).

1 FOUNDATIONS OF REACTION KINETICS

Imagine you are a farmer in the Northwest Region watching a bag of NPK fertilizer dissolve in a water tank for irrigation. The speed at which the nutrients become available determines the health of your crops. Understanding why some reactions are fast and others slow, and how to measure and control them, is the power of reaction kinetics.

1.1 What Is Reaction Rate?

Chemical reactions do not happen instantaneously; they proceed at a certain speed. The **rate of reaction** tells us how quickly reactants are consumed or products are formed per unit time.

Consider a general reaction:

$$aA + bB \rightarrow cC + dD$$

where a, b, c, d are the stoichiometric coefficients. The rate of reaction v (or r) is defined so that it has the same value whether we follow the disappearance of a reactant or the appearance of a product.

DEFINITION **Rate of Reaction**

For the reaction $aA + bB \rightarrow cC + dD$, the rate of reaction v is defined as:

$$v = -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = +\frac{1}{c}\frac{d[C]}{dt} = +\frac{1}{d}\frac{d[D]}{dt}$$

where [X] denotes the concentration of species X in mol dm^{-3}, and t is time in seconds. The negative sign for reactants ensures v is a positive quantity.

PROPERTY **Rate of Change of Concentration vs. Reaction Rate**

- For a reactant: rate of disappearance (rate of change of concentration) $= -\frac{d[\text{reactant}]}{dt} = a \times v$.
- For a product: rate of appearance (rate of change of concentration) $= +\frac{d[\text{product}]}{dt} = c \times v$.

The **units of reaction rate** v are always $\text{mol dm}^{-3}\,\text{s}^{-1}$ (or M s^{-1}). The units of $\frac{d[X]}{dt}$ are also $\text{mol dm}^{-3}\,\text{s}^{-1}$.

1.2 Average, Instantaneous, and Initial Rates

EXEMPLE **Decomposition of Dinitrogen Pentoxide**

The gas-phase decomposition $2\,N_2O_5(g) \longrightarrow 4\,NO_2(g) + O_2(g)$ is studied. At $t = 0$, $[N_2O_5] = 0.100\ \text{mol dm}^{-3}$. After 100 s, $[N_2O_5] = 0.070\ \text{mol dm}^{-3}$.

1. Calculate the average rate of reaction over the first 100 s.
2. Express the rate in terms of NO_2 formation.

Solution:

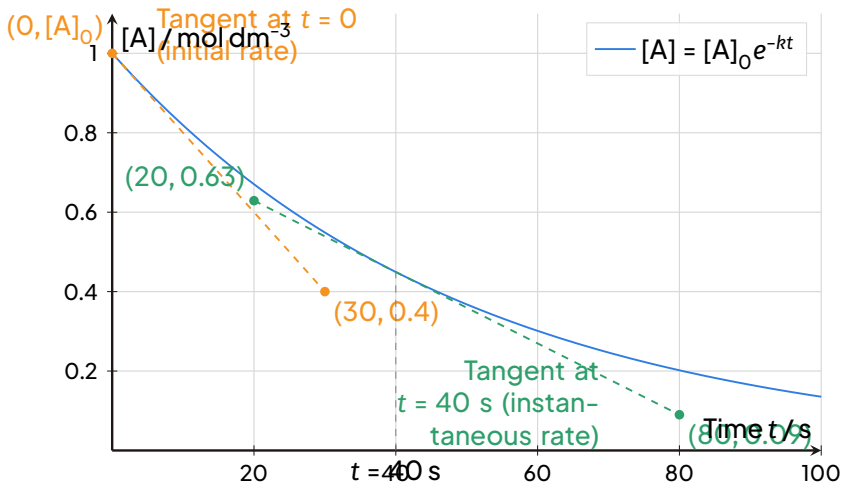
1. Change in $[N_2O_5] = 0.070 - 0.100 = -0.030\ \text{mol dm}^{-3}$. Average rate of disappearance of $N_2O_5 = \frac{0.030}{100} = 3.0 \times 10^{-4}\ \text{mol dm}^{-3}\ \text{s}^{-1}$. Reaction rate $v = -\frac{1}{2}\frac{\Delta[N_2O_5]}{\Delta t} = \frac{1}{2} \times 3.0 \times 10^{-4} = \mathbf{1.5 \times 10^{-4}}\ \text{mol dm}^{-3}\ \text{s}^{-1}$.
2. $v = +\frac{1}{4}\frac{\Delta[NO_2]}{\Delta t} \Rightarrow \frac{\Delta[NO_2]}{\Delta t} = 4v = 6.0 \times 10^{-4}\ \text{mol dm}^{-3}\ \text{s}^{-1}$.

The **average rate** over a time interval Δt is $\frac{\Delta[\text{species}]}{\Delta t}$ (with stoichiometric factor). However, the rate usually changes as concentrations change. We need the **instantaneous rate** at a specific time t, which is the derivative $\frac{d[\text{species}]}{dt}$ at that instant.

METHOD **Finding Instantaneous Rate from a Concentration–Time Graph**

1. Plot concentration of a reactant or product against time.
2. At the desired time t, draw a tangent to the curve.
3. Choose two points on the tangent line, (t_1, c_1) and (t_2, c_2), far apart for accuracy.
4. Calculate the gradient of the tangent: gradient $= \frac{c_2 - c_1}{t_2 - t_1} = \frac{d[\text{species}]}{dt}$.
5. Apply the stoichiometric factor to obtain the reaction rate v:
   - If the curve is for a reactant A (coefficient a): gradient is negative. $v = -\frac{1}{a} \times$ gradient.
   - If the curve is for a product C (coefficient c): gradient is positive. $v = +\frac{1}{c} \times$ gradient.

The **initial rate** is the instantaneous rate at $t = 0$. It is especially important because it depends only on the initial concentrations, before any significant product has formed or reactant has been depleted.



Concentration–time curve for a reactant A showing the tangent at $t = 0$ (initial rate) and at $t = 40$ s (instantaneous rate). The gradient of each tangent gives the rate of change of concentration at that instant.

WARNING **Average vs. Instantaneous Rate**

Common mistake: Using $\frac{\Delta[A]}{\Delta t}$ over a large interval (e.g., 0 to 100 s) and calling it the rate at $t = 50$ s. This gives the average rate over that interval, not the instantaneous rate at the midpoint. The instantaneous rate is the gradient of the tangent at the point. For a curved graph, the average rate over an interval is not equal to the instantaneous rate at the midpoint (except for a straight line, i.e., zero-order reactions).

1.3 Factors Affecting the Rate of Reaction

Why does fertilizer dissolve quickly in hot water but slowly in cold? Why does powdered limestone react faster with acid than a large chip? The main factors are:

1. **Concentration (or pressure for gases):** Higher concentration → more particles per unit volume → more frequent collisions → higher rate. For gases, increasing pressure increases concentration ($c = n/V = p/RT$).
2. **Temperature:** Increasing temperature increases the kinetic energy of molecules. More molecules have energy ≥ E_a (activation energy), and collisions are more frequent. A rough rule: rate ≈ doubles for every 10 K rise.
3. **Surface Area (heterogeneous reactions):** For a solid reacting with a liquid or gas, only the surface is available. Powdering increases surface area → more collision sites → faster rate.
4. **Catalyst:** A substance that increases the rate without being consumed. It provides an alternative pathway with **lower activation energy** E_a. It does not change ΔH or the equilibrium constant.

5. **Nature of Reactants:** Ionic reactions (e.g., $Ag^+ + Cl^- \longrightarrow AgCl\downarrow$) are almost instantaneous. Covalent bond breaking (e.g., $H_2 + Cl_2 \longrightarrow 2\,HCl$) requires energy input and is slower.

DID YOU KNOW? **Cameroon Context: Limestone and Cement**

In Figuil (North Region), limestone ($CaCO_3$) is heated to make cement. The decomposition $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$ is slow at room temperature but rapid at 900–1000 °C. The high temperature supplies the activation energy to break strong Ca–O and C–O bonds. In the lab, we study the reverse: $CaCO_3 + 2\,HCl \longrightarrow CaCl_2 + H_2O + CO_2$. Using powdered limestone instead of chips dramatically increases the rate — a classic demonstration of surface area effect.

1.4 Why Study Reaction Kinetics?

- **Industrial Optimisation:** Haber process ($N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3$) — kinetics tells us the best T, p, and catalyst (Fe) to maximise yield per unit time. In Cameroon, SONARA (National Oil Refinery) uses kinetic data to optimise cracking and reforming units.
- **Environmental Monitoring:** Rate of SO_2 oxidation to SO_3 (acid rain), NO_x formation in engines, CFC decomposition in stratosphere. Predicting pollutant lifetimes requires rate constants.
- **Biological Processes:** Enzyme kinetics (Michaelis–Menten) governs digestion, respiration, drug metabolism. Understanding rates helps design medicines (e.g., antiretrovirals for HIV).
- **Food Preservation:** Rates of spoilage reactions (oxidation, microbial growth) determine shelf life. Refrigeration lowers T → slows rates.
- **Safety:** Explosions are extremely fast reactions. Kinetics helps design controlled detonations (mining in East Region) and prevent accidental runaway reactions.

1.5 Worked Example: Rates from a Graph

WORKED EXAMPLE 1 **Determining Rates from a Concentration–Time Graph**

For the reaction $2\,A \longrightarrow B + C$, the concentration of A was measured over time:

t/s	0	20	40	60	80	100
[A]/mol dm^{-3}	1.00	0.67	0.45	0.30	0.20	0.13

1. Plot [A] vs t and draw a smooth curve.
2. Determine the initial rate of reaction.
3. Determine the instantaneous rate at t = 40 s.
4. Calculate the average rate of formation of B between t = 20 s and t = 60 s.

SOLUTION

Solution:

1. (Graph plotted — see figure above for similar shape.)
2. Initial rate: Draw tangent at $t = 0$. Choose points $(0, 1.00)$ and $(20, 0.60)$ on tangent. Gradient $= \frac{0.60-1.00}{20-0} = -0.020\ \text{mol dm}^{-3}\ \text{s}^{-1}$. Rate of disappearance of A $= 0.020\ \text{mol dm}^{-3}\ \text{s}^{-1}$. Reaction rate $v = -\frac{1}{2}\frac{d[\text{A}]}{dt} = \frac{1}{2} \times 0.020 = \mathbf{0.010}\ \text{mol dm}^{-3}\ \text{s}^{-1}$.
3. At $t = 40$ s: $[\text{A}] \approx 0.45$. Tangent gradient $\approx \frac{0.30-0.60}{60-20} = -0.0075\ \text{mol dm}^{-3}\ \text{s}^{-1}$. $v = \frac{1}{2} \times 0.0075 = \mathbf{3.75 \times 10^{-3}}\ \text{mol dm}^{-3}\ \text{s}^{-1}$.
4. Average rate of formation of B: $\Delta[\text{A}] = 0.30 - 0.67 = -0.37\ \text{mol dm}^{-3}$ over $\Delta t = 40$ s. Average rate of disappearance of A $= \frac{0.37}{40} = 9.25 \times 10^{-3}\ \text{mol dm}^{-3}\ \text{s}^{-1}$. Average reaction rate $v_{\text{av}} = \frac{1}{2} \times 9.25 \times 10^{-3} = 4.625 \times 10^{-3}\ \text{mol dm}^{-3}\ \text{s}^{-1}$. Rate of formation of B $= +\frac{d[\text{B}]}{dt} = 1 \times v_{\text{av}} = \mathbf{4.6 \times 10^{-3}}\ \text{mol dm}^{-3}\ \text{s}^{-1}$ (2 s.f.).

1.6 Summary of Key Points

KEY POINTS **Foundations of Reaction Kinetics**

- Rate of reaction $v = \frac{1}{\nu_i}\frac{d[\text{C}_i]}{dt}$ (positive for products, negative for reactants). Units: $\text{mol dm}^{-3}\ \text{s}^{-1}$.
- Average rate over Δt: $\frac{\Delta[\text{species}]}{\Delta t}$ (with stoichiometric factor).
- Instantaneous rate at t: gradient of tangent to concentration–time curve at t.
- Initial rate: instantaneous rate at $t = 0$; depends only on initial concentrations.
- Factors increasing rate: higher concentration/pressure, higher temperature, larger surface area (heterogeneous), catalyst (lowers E_a).
- Applications: industrial synthesis, environmental chemistry, biochemistry, food technology, safety.

EXERCISE 1 **Practice Questions**

1. For the reaction $\text{N}_2(\text{g}) + 3\,\text{H}_2(\text{g}) \longrightarrow 2\,\text{NH}_3(\text{g})$, if the rate of formation of NH_3 is $2.0 \times 10^{-4}\ \text{mol dm}^{-3}\ \text{s}^{-1}$, what are the rates of disappearance of N_2 and H_2?

2. A student measures the volume of CO_2 evolved when $CaCO_3$ reacts with excess HCl. Explain why the rate of reaction decreases with time even though the temperature is constant.
3. Sketch a concentration–time curve for a reactant in a first-order reaction. On the same axes, show how the initial rate and the rate at half-life would be determined graphically.
4. In the Haber process, the reaction is carried out at 450 °C and 200 atm with an iron catalyst. Explain, using kinetic theory, the effect of each of these three conditions on the rate of reaction.

SOLUTION **Exercise 1**

1. $v = +\frac{1}{2}\frac{d[NH_3]}{dt} = 1.0 \times 10^{-4}\ \text{mol dm}^{-3}\,\text{s}^{-1}$. Rate of disappearance of $N_2 = -\frac{d[N_2]}{dt} = 1 \times v = \mathbf{1.0 \times 10^{-4}}\ \text{mol dm}^{-3}\,\text{s}^{-1}$. Rate of disappearance of $H_2 = -\frac{d[H_2]}{dt} = 3 \times v = \mathbf{3.0 \times 10^{-4}}\ \text{mol dm}^{-3}\,\text{s}^{-1}$.
2. As HCl is consumed, its concentration decreases. Fewer H^+ ions per unit volume → fewer effective collisions per second with the $CaCO_3$ surface → rate decreases.
3. Curve: exponential decay. Tangent at $t = 0$ (steepest). Tangent at $t = t_{1/2}$ where $[A] = [A]_0/2$ (less steep).
4. 450 °C: high T → more molecules with $E \geq E_a$ → higher rate (but compromise with equilibrium yield). 200 atm: high pressure → high concentration of gases → more frequent collisions → higher rate. Fe catalyst: provides alternative pathway with lower E_a → higher rate at same T.

2 CHEMICAL METHOD – TITRIMETRIC RATE DETERMINATION

In the previous section, we established the foundations of reaction kinetics: we defined the rate of reaction, explored the factors that influence it (concentration, temperature, catalysts, surface area), and learned how to express rate mathematically. We also saw that to determine a rate law experimentally, we need reliable concentration–time data. But how do we actually obtain that data for a reaction happening in solution?

Imagine you are studying the hydrolysis of an ester, $CH_3COOCH_2CH_3 + H_2O \longrightarrow CH_3COOH + CH_3CH_2OH$, catalysed by acid. The reaction proceeds slowly enough to be followed over minutes or hours, but you cannot "see" the concentration changing directly. You need a way to "freeze" the reaction at specific instants, analyse the composition, and then plot concentra-

tion versus time. This is the essence of the **chemical (titrimetric) method** — a classic, robust technique that remains a cornerstone of kinetic studies in our laboratories.

2.1 Principle: Quenching and Aliquot Analysis

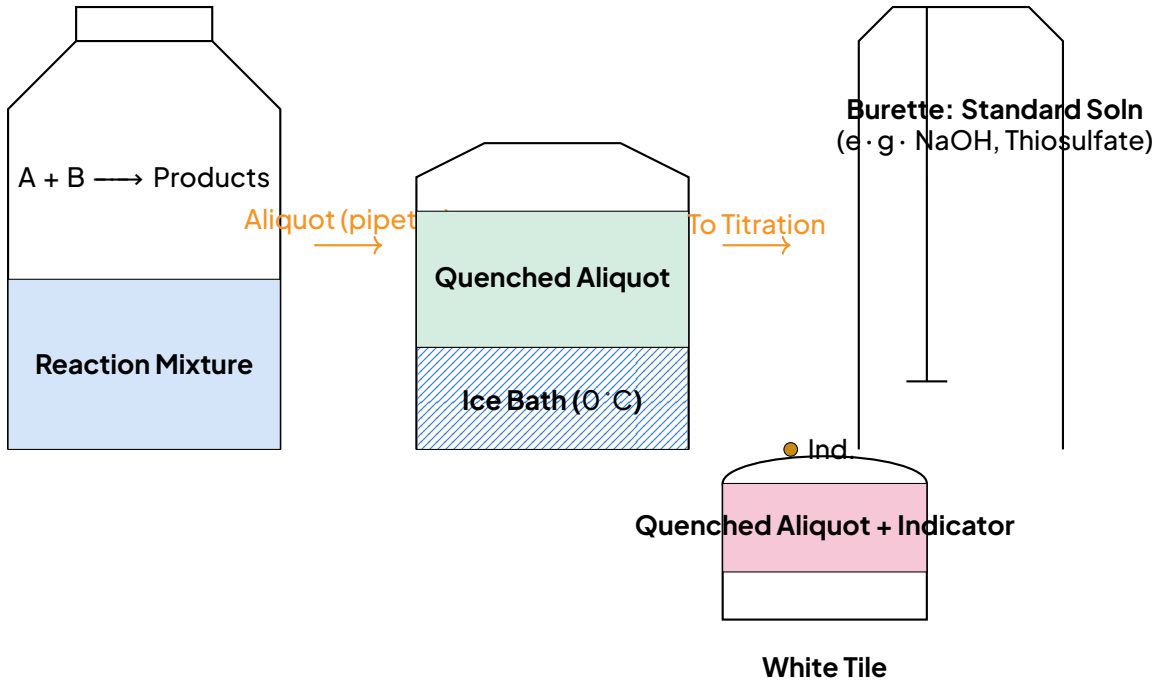
The titrimetric method relies on a simple but powerful idea: **remove a small portion (aliquot) of the reaction mixture at a known time, stop the reaction instantly in that portion, and then determine the concentration of a reactant or product by titration.**

DEFINITION **Quenching**

Quenching is the process of rapidly stopping (or drastically slowing down) a chemical reaction in a sampled aliquot so that its composition remains essentially unchanged from the moment of sampling until analysis. Common quenching techniques include:

- Cooling: Plunging the aliquot into an ice–water bath (0 °C) to reduce the rate constant k to near zero.
- Dilution: Adding the aliquot to a large volume of cold water or solvent, reducing concentrations and thus the rate.
- Neutralisation / Reagent addition: For acid-catalysed reactions, adding excess standard base to neutralise the catalyst; for reactions involving a specific reagent, adding a scavenger that consumes a reactant instantly.

The key requirement: quenching must be effectively instantaneous compared to the reaction half-life. If the reaction continues during quenching, the measured concentration will not correspond to the true concentration at the sampling time t.



Schematic of the quenching–titration workflow: (1) Reaction flask maintained at constant temperature; (2) Aliquot withdrawn by pipette and immediately quenched in ice bath; (3) Quenched aliquot titrated against a standard solution with a suitable indicator.

2.2 Experimental Procedure: Step by Step

A typical experiment follows this rigorous protocol:

METHOD **Titrimetric Rate Determination Protocol**

1. Prepare reactants: Make up solutions of known initial concentrations. If temperature dependence is studied, thermostat the reactants separately at the chosen temperature (e.g., 25.0 °C) in a water bath.
2. Initiate reaction: Mix reactants in the reaction flask (start the clock, $t = 0$). Swirl gently to ensure homogeneity. Maintain temperature control.
3. Sampling (Aliquot withdrawal): At predetermined times $t_1, t_2, t_3, \ldots$, use a clean, dry pipette (often 10.0 cm^3 or 25.0 cm^3) to withdraw an aliquot. Immediately transfer it to the quenching vessel.
4. Quenching: Ensure instantaneous quenching. For an ice bath, the quenching vessel should be pre-cooled and contain enough ice-water slurry to bring the aliquot to 0 °C within seconds.
5. Titration: Transfer the quenched aliquot (or titrate directly in the quenching vessel

if volume permits) to a conical flask. Add 2–3 drops of a suitable indicator. Titrate against a standard solution of known concentration C_{std} until the endpoint.

6. Record data: Note the time t and the titre volume V_{std} for each aliquot. Also perform a "$t = \infty$" titration (after reaction is complete, e.g., 10 half-lives) to find the final concentration.
7. Repeat: Carry out at least two runs for reproducibility.

2.3 Stoichiometric Calculations: From Titre to Concentration

The raw data are times t and titre volumes V_{std}. We must convert these into concentrations $[A]_t$ of the species being monitored (reactant or product).

METHOD **Calculating Concentration from Titration Data**

Let the reaction be:

$$aA + bB \rightarrow \text{products}$$

Suppose we titrate species A (reactant) with standard titrant T of concentration C_{std}. The titration reaction is: $p\,A + q\,T \longrightarrow$ products.

1. Moles of titrant used at time t: $n_T = C_{std} \times V_{std}(t)$ (in dm^3).
2. Moles of A in the aliquot: $n_A(t) = \frac{p}{q} \times n_T = \frac{p}{q} C_{std} V_{std}(t)$.
3. Concentration of A in the original reaction mixture at time t:

$$[A]_t = \frac{n_A(t)}{V_{aliquot}} = \frac{p}{q} \frac{C_{std} V_{std}(t)}{V_{aliquot}}$$

where $V_{aliquot}$ is the pipette volume (in dm^3).

If titrating a product P formed with stoichiometry $aA \rightarrow rP$, then $[A]_t = [A]_0 - \frac{a}{r}[P]_t$.

PROPERTY **Concentration–Time Table Construction**

After calculating $[A]_t$ for each time point, construct a table:

Time t / s	Titre V_{std} / cm^3	$[A]_t$ / mol dm^{-3}	$\ln[A]_t$ or $1/[A]_t$
0	V_0	$[A]_0$	$\ln[A]_0$
t_1	V_1	$[A]_1$	$\ln[A]_1$
t_2	V_2	$[A]_2$	$\ln[A]_2$
⋮	⋮	⋮	⋮
∞	V_∞	$[A]_\infty$ (often 0)	–

This table is the raw material for graphical determination of order (Section 4).

2.4 Choice of Indicator and Standard Solution

The success of the method hinges on a sharp, reproducible endpoint.

- **Acid–base titrations:** Phenolphthalein (pH 8.2–10, colourless to pink) for strong base vs weak/strong acid; Methyl orange (pH 3.1–4.4, red to yellow) for strong acid vs weak base. **Warning:** In kinetic runs where pH changes during reaction (e.g., acid-catalysed ester hydrolysis producing acid), the indicator must be chosen for the final pH of the quenched aliquot.
- **Redox titrations:** Starch indicator (blue-black complex with I_2) for thiosulfate–iodine titrations ($I_2 + 2\,S_2O_3^{2-} \longrightarrow 2\,I^- + S_4O_6^{2-}$). Add starch near the endpoint (pale yellow) to avoid adsorption errors.
- **Standard solutions:** Must be primary standards (stable, high purity, non-hygroscopic) or standardised against one. Examples: Na_2CO_3 (acid–base), $K_2Cr_2O_7$ or $Na_2S_2O_3$ (redox). Concentration known to ±0.1%.

2.5 Worked Example: Acid-Catalysed Hydrolysis of Ethyl Ethanoate

WORKED EXAMPLE 2 **Determining Rate Constant from Titration Data**

Statement: The acid-catalysed hydrolysis of ethyl ethanoate is studied at 25 °C: $CH_3COOCH_2CH_3(l) + H_2O(l) \xrightarrow{H^+} CH_3COOH(aq) + CH_3CH_2OH(aq)$. A reaction mixture is prepared by mixing 50.0 cm^3 of 0.200 mol dm^{-3} ester with 50.0 cm^3 of 0.100 mol dm^{-3} HCl (catalyst). At various times, 10.0 cm^3 aliquots are withdrawn, quenched in ice, and titrated against 0.100 mol dm^{-3} NaOH using phenolphthalein. The titres are:

t / min	0	10	20	30	∞
V_{NaOH} / cm^3	15.0	20.5	24.8	28.2	40.0

1. Explain why the t = 0 titre is not zero.
2. Calculate $[CH_3COOH]_t$ at each time.

3. Show that the reaction is first order with respect to ester and determine the rate constant k (in s^{-1}).

SOLUTION

Detailed Solution:

1. Why $V_0 \neq 0$? At $t = 0$, the mixture already contains the catalyst HCl (0.100 mol dm^{-3} before mixing, 0.050 mol dm^{-3} after mixing 50+50 cm^3). The titration with NaOH neutralises all free acid present: the catalyst HCl plus any acetic acid formed. So V_0 corresponds to the initial catalyst concentration.

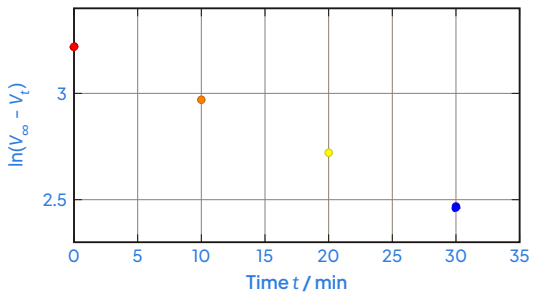
2. Standard Approach for Ester Hydrolysis. The titration measures total acid (HCl + CH_3COOH). Both react 1:1 with NaOH. Let $V_{catalyst} = V_0$ be the volume of NaOH needed to neutralise HCl alone in the 10.0 cm^3 aliquot. Let V_∞ be the titre at completion (all ester hydrolysed). Then:

- Volume of NaOH proportional to acetic acid formed at time $t = V_t - V_0$.
- Volume of NaOH proportional to ester remaining at time $t = V_\infty - V_t$.

Since $[Ester]_t \propto (V_\infty - V_t)$, we can use $(V_\infty - V_t)$ directly in the first-order integrated rate law.

t / min	V_t / cm^3	$V_\infty - V_t$ (prop. to [Ester])	$\ln(V_\infty - V_t)$
0	15.0	25.0	3.219
10	20.5	19.5	2.970
20	24.8	15.2	2.721
30	28.2	11.8	2.468

3. Graphical Test for First Order. For a first-order reaction: $\ln[A]_t = \ln[A]_0 - kt$. Since $[A]_t \propto (V_\infty - V_t)$, a plot of $\ln(V_\infty - V_t)$ vs t should be linear with slope $-k$.



The points fall on a straight line → First order confirmed.

Slope calculation: Using $t = 0$ and $t = 30$ min: Slope = $\frac{2.468-3.219}{30-0} = \frac{-0.751}{30} = -0.02503\ min^{-1}$.

$k = 0.0250\ min^{-1}$. Convert to s^{-1}: $k = \frac{0.0250}{60} = \mathbf{4.17 \times 10^{-4}}\ s^{-1}$. (Using linear regression on all points gives $k \approx 4.2 \times 10^{-4}\ s^{-1}$).

2.6 Sources of Error and Precautions

Even with careful technique, the titrimetric method has inherent limitations. Understanding these is crucial for the Planning and Analysis papers.

WARNING **Common Pitfalls in Titrimetric Kinetics**

1. Incomplete / Slow Quenching: If the aliquot takes 30 seconds to cool to 0 °C, and the half-life is 2 minutes, significant reaction occurs during quenching. The measured $[A]_t$ will be lower than the true value (for a reactant). Remedy: Use a large ice bath, small aliquot volume, or chemical quenching (e.g., adding excess base to acid-catalysed reaction).
2. Sampling Volume Error: Using a 25 cm^3 pipette removes a significant fraction of the reaction mixture, changing the total volume and concentrations over the course of the experiment. Remedy: Use a 10 cm^3 pipette; keep total volume large (e.g., 250 cm^3); or correct mathematically for volume loss.
3. Endpoint Detection: Phenolphthalein endpoint is subtle (colourless to faint pink). Over-titration adds excess NaOH, overestimating acid concentration. Remedy: Use a white tile, good lighting, and titrate dropwise near the end. Carry out a "blank" titration of the indicator in water to correct for indicator blank.
4. Indicator Interference: In redox kinetics (e.g., $S_2O_8^{2-}/I^-$ with thiosulfate), starch must be added only when the solution is pale yellow (low I_2). Adding it too early forms a stable starch–iodine complex that is hard to titrate.
5. Standard Solution Stability: NaOH absorbs CO_2 from air, forming Na_2CO_3, which changes its concentration and gives a double endpoint with phenolphthalein/methyl orange. Remedy: Standardise NaOH frequently against KHP (potassium hydrogen phthalate); store in a bottle with a CO_2 guard tube (soda lime).

2.7 Advantages and Limitations

Aspect	Advantages	Limitations
Equipment	Simple, inexpensive glassware (burette, pipette, flask). Available in every school lab.	Labour-intensive; manual sampling limits time resolution (cannot follow very fast reactions < 1 min).
Versatility	Works for any reaction where a reactant or product can be titrated (acid–base, redox, complexometric, precipitation).	Requires a suitable titration reaction; not universal (e.g., no simple titration for N_2O_5 decomposition).
Accuracy	High precision possible (±0.1%) with good technique and primary standards.	Cumulative errors: sampling + quenching + titration + indicator.
Data Quality	Gives absolute concentrations directly (mol dm^{-3}), not just relative signals.	Discrete data points only; cannot capture continuous concentration profile.

2.8 Connection to Rate Law Determination

The concentration–time data (t, $[A]_t$) obtained here are the **raw material** for the next section. We will use them to:

- Plot $[A]_t$ vs t → tangent method for instantaneous rates.
- Plot $\ln[A]_t$ vs t and $1/[A]_t$ vs t → graphical determination of order (Section 4, Lesson 97).
- Use initial rates from different initial concentrations → initial rate method (Section 4, Lesson 96).

KEY POINTS **Key Points for Examination**

- Quenching must be instantaneous (ice bath, dilution, neutralisation). Incomplete quenching → systematic error (rate appears slower).
- Concentration calculation: $[A]_t = \frac{\text{stoich factor} \times C_{std} \times V_{std}(t)}{V_{aliquot}}$. For ester hydrolysis: $[\text{Ester}]_t \propto (V_\infty - V_t)$.
- First-order test: $\ln(V_\infty - V_t)$ vs t is linear; slope = $-k$.
- Major errors: Volume change due to sampling, indicator blank, CO_2 absorption by NaOH, slow quenching.
- Units of k: First order → s^{-1}; Second order → $dm^3\ mol^{-1}\ s^{-1}$.

DID YOU KNOW? **Titrimetric Kinetics in Cameroon's Industries**

In the CDC (Cameroon Development Corporation) rubber and palm oil processing plants,

the saponification value of oils (a measure of average molecular weight) is determined by refluxing oil with excess ethanolic KOH and back-titrating the remaining KOH with HCl. While this is an analytical method, the kinetics of saponification (rate of hydrolysis of triglycerides) is crucial for designing the reactors: too fast → runaway temperature; too slow → low throughput. Chemical engineers use pilot-plant titration data (exactly as described here) to size industrial saponification reactors for soap production in Douala and Bassa.

EXERCISE 2 **Practice Problem**

The decomposition of hydrogen peroxide catalysed by MnO_2 is studied: $2\,H_2O_2(aq) \longrightarrow 2\,H_2O(l) + O_2(g)$. At fixed intervals, 10.0 cm^3 aliquots are withdrawn, the MnO_2 is filtered off quickly, and the filtrate is titrated with 0.0200 mol dm^{-3} $KMnO_4$ in acidic medium: $2\,MnO_4^- + 5\,H_2O_2 + 6\,H^+ \longrightarrow 2\,Mn^{2+} + 5\,O_2 + 8\,H_2O$. The following titres are obtained:

t / min	0	5	10	15	20
V_{KMnO_4} / cm^3	25.0	18.4	13.6	10.0	7.4

1. Calculate $[H_2O_2]_t$ at each time.
2. Plot $\ln[H_2O_2]_t$ vs t and determine the order and rate constant k.
3. Suggest a better quenching method than filtration for this heterogeneous catalysis.

SOLUTION **Exercise 2**

1. Stoichiometry: $2\,MnO_4^- \equiv 5\,H_2O_2$. Moles $H_2O_2 = \frac{5}{2} \times C_{KMnO_4} \times V_{KMnO_4}$. $[H_2O_2]_t = \frac{5}{2} \times \frac{0.0200 \times V_{KMnO_4}(\text{in dm}^3)}{0.0100} = 5.00 \times V_{KMnO_4}(\text{in dm}^3)$ mol dm^{-3}. V in cm^3 → $[H_2O_2] = 0.00500 \times V_{KMnO_4}$. t = 0: 0.125 M; t = 5: 0.0920 M; t = 10: 0.0680 M; t = 15: 0.0500 M; t = 20: 0.0370 M.
2. $\ln[H_2O_2]$: −2.08, −2.38, −2.69, −2.99, −3.29. Plot is linear → First order. Slope ≈ $\frac{-3.29-(-2.08)}{20} = -0.0605\ min^{-1} = 1.01 \times 10^{-3}\ s^{-1}$.
3. Filtration is slow. Better: Add a complexing agent (e.g., EDTA or phosphate buffer) that instantly binds Mn^{2+}/poisons the catalyst surface, or dilute massively into cold acidified solution containing a MnO_2 inhibitor.

3 PHYSICAL METHODS OF RATE MEASUREMENT

In the previous section we followed a reaction by the chemical method: we withdrew samples at known times, quenched them, and titrated. That method is reliable but slow, destructive (each sample is lost), and impossible for very fast reactions. **Physical methods** overcome these drawbacks: instead of removing samples, we continuously monitor some physical property of the reaction mixture that changes in proportion to the concentration of a reactant or product — its colour, its electrical conductivity, its volume, or the pressure/volume of a gas it produces. The mixture is never disturbed, readings can be taken every few seconds, and a complete rate curve is obtained from a single experiment.

The general strategy is always the same:

1. choose a measurable physical quantity X that is a known, simple function of a concentration (ideally proportional to it);
2. calibrate, i.e. establish the relation $X = f(c)$ using standard solutions;
3. record X against time during the reaction;
4. convert $X(t)$ into $c(t)$, and hence obtain the rate as the gradient of the concentration–time curve, exactly as in Section 2.

3.1 Colorimetry

Intuition.

Many reactions involve a coloured species. When purple acidified permanganate oxidises ethanedioic acid,

$$2\,MnO_4^-(aq) + 5\,C_2O_4^{2-}(aq) + 16\,H^+(aq) \longrightarrow 2\,Mn^{2+}(aq) + 10\,CO_2(g) + 8\,H_2O(l),$$

the solution slowly turns from purple to colourless: the intensity of the colour is a direct, visible measure of $[MnO_4^-]$. A **colorimeter** (or spectrophotometer) simply makes this judgement quantitative: it shines light of a chosen wavelength through the solution and measures how much is absorbed.

DEFINITION **Beer–Lambert law**

When monochromatic light passes through a solution of an absorbing species, the **absorbance** A is proportional to the concentration c of the absorbing species and to the path length l of the cell:

$$A = \varepsilon\, l\, c$$

where ε is the **molar absorption coefficient** (units $dm^3\,mol^{-1}\,cm^{-1}$), l is in cm and c in $mol\,dm^{-3}$. Absorbance itself is dimensionless: $A = \log_{10}(I_0/I)$, where I_0 and I are the intensities of the incident and transmitted light.

PROPERTY **Consequence for kinetics**

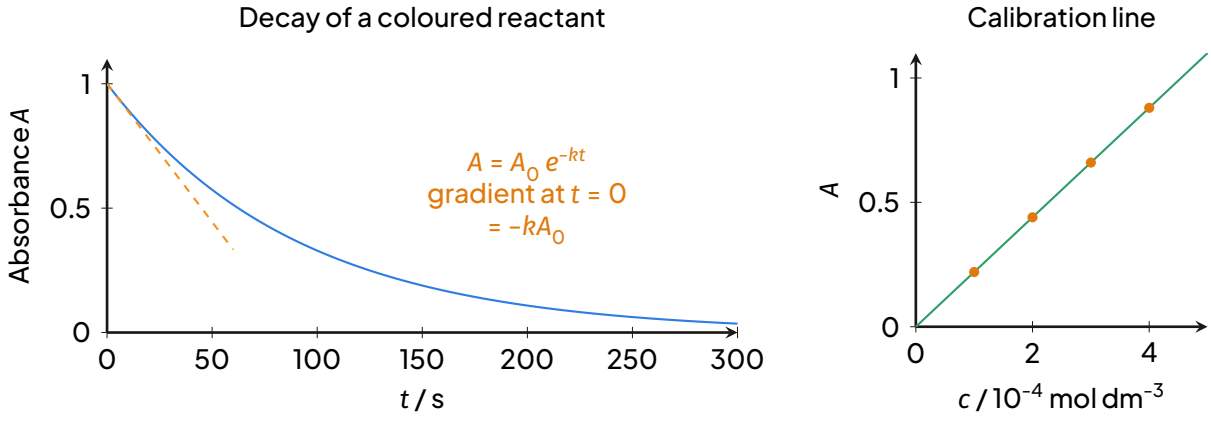
Since ε and l are constant for a given experiment, $A \propto c$. Therefore a plot of A against time has exactly the same shape as the concentration–time curve, and

$$\text{rate} = -\frac{dc}{dt} = -\frac{1}{\varepsilon l}\frac{dA}{dt}.$$

In practice one converts A to c with a **calibration curve**: the absorbances of several standard solutions of known concentration are measured, and A is plotted against c. The result should be a straight line through the origin, from which any measured A can be read off as a concentration.

WARNING **Choosing the wavelength**

Colorimetry requires that only one species absorbs significantly at the chosen wavelength. If reactant and product absorb at the same wavelength (overlapping spectra), the measured absorbance no longer tracks a single concentration and the kinetic analysis is wrong. Always choose the wavelength of maximum absorption of the species of interest (λ_{max}), and check that other species are transparent there. Also remember that the Beer–Lambert law fails at high concentrations (typically $c > 0.01\ \text{mol dm}^{-3}$): work in the linear range of the calibration curve.



Colorimetric following of a reaction. Left: absorbance of the coloured reactant falls exponentially with time; the initial rate is obtained from the tangent at $t = 0$. Right: the calibration line $A = \varepsilon l c$ used to convert absorbance into concentration.

WORKED EXAMPLE 3 **Using a calibration curve**

The hydrolysis of a coloured dye is followed colorimetrically at its λ_{max}. A calibration experiment gives $\varepsilon l = 1.1 \times 10^4\ \text{dm}^3\ \text{mol}^{-1}$. During the reaction the absorbance falls from $A_0 = 0.88$ at $t = 0$ to $A = 0.44$ at $t = 120$ s. Find (a) the initial concentration of the dye, (b) the average rate of disappearance over the first 120 s.

SOLUTION

(a) From $A = \varepsilon l c$:

$$c_0 = \frac{A_0}{\varepsilon l} = \frac{0.88}{1.1 \times 10^4} = 8.0 \times 10^{-5}\ \text{mol dm}^{-3}.$$

(b) At $t = 120$ s: $c = 0.44/(1.1 \times 10^4) = 4.0 \times 10^{-5}\ \text{mol dm}^{-3}$.

$$\text{average rate} = -\frac{\Delta c}{\Delta t} = -\frac{(4.0 - 8.0) \times 10^{-5}}{120} = 3.3 \times 10^{-7}\ \text{mol dm}^{-3}\ \text{s}^{-1}.$$

Note that the absorbance halved in 120 s; because $A \propto c$, the concentration also halved — a hint that the reaction may be first order (constant half-life), to be confirmed in Section 4.

3.2 Conductimetry

Intuition.

A solution conducts electricity through its ions. If a reaction produces or consumes ions, the conductivity of the mixture changes as the reaction proceeds, and two inert electrodes (usually platinum) dipping into the flask give us a continuous, non-destructive concentration probe. A classic example is the alkaline hydrolysis of an ester:

$$CH_3COOC_2H_5(aq) + OH^-(aq) \longrightarrow CH_3COO^-(aq) + C_2H_5OH(aq).$$

Here the highly mobile OH^- ion is replaced by the much less mobile CH_3COO^- ion, so the conductivity falls steadily during the reaction.

DEFINITION **Conductivity and molar ionic conductivities**

The **conductivity** κ (unit S cm^{-1} or S m^{-1}) of an ionic solution is the sum of the contributions of all ions present:

$$\kappa = \sum_i \lambda_i c_i$$

where c_i is the concentration of ion i and λ_i its **molar ionic conductivity** (unit $\text{S cm}^2\ \text{mol}^{-1}$). Ions such as H^+ ($\lambda \approx 350\ \text{S cm}^2\ \text{mol}^{-1}$) and OH^- ($\lambda \approx 198\ \text{S cm}^2\ \text{mol}^{-1}$) have exceptionally large λ values, so reactions involving them give large, easily measured conductivity changes.

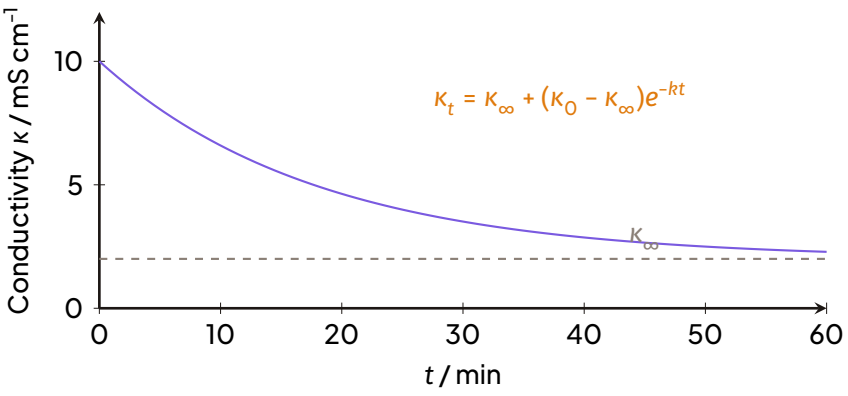
METHOD **From conductivity to concentration**

To convert a conductivity reading κ_t into a concentration:

1. Identify all ions present at time t and write $\kappa_t = \sum_i \lambda_i c_i$.
2. Use the stoichiometry of the reaction to express every c_i in terms of a single variable (e.g. the extent of reaction x or the concentration of the limiting reactant).
3. Use the initial and final readings κ_0 and κ_∞ as anchors. If the conductivity change is due solely to the consumption of one ionic reactant (others constant), the fraction of that reactant remaining is
$$\frac{[\text{reactant}]_t}{[\text{reactant}]_0} = \frac{\kappa_t - \kappa_\infty}{\kappa_0 - \kappa_\infty}.$$
4. Plot concentration against time and take gradients to obtain rates.

EXEMPLE **Ester hydrolysis followed conductimetrically**

For $CH_3COOC_2H_5 + OH^- \longrightarrow CH_3COO^- + C_2H_5OH$ with the ester in large excess (pseudo-first-order conditions): at $t = 0$ the conductivity is due essentially to Na^+ (from NaOH) and OH^-; at $t = \infty$ all OH^- has been replaced by CH_3COO^-. Since $\lambda(OH^-) > \lambda(CH_3COO^-)$, κ decreases linearly with the extent of reaction, and $[OH^-]_t \propto (\kappa_t - \kappa_\infty)$.



Conductivity–time trace for the alkaline hydrolysis of ethyl ethanoate: the fast OH^- ions are replaced by slower CH_3COO^- ions, so κ falls towards its final value κ_∞.

WARNING **Conditions for valid conductimetry**

- Keep the temperature constant (conductivity increases roughly 2% per degree Celsius).
- Use an AC supply (typically 1 kHz) to avoid electrolysis and polarisation at the electrodes.
- Ensure the conductivity change is dominated by one ionic transformation; reactions between neutral molecules with no ionic change cannot be followed this way.
- The cell constant must be known or calibrated with a standard solution (e.g. 0.01 M KCl).

3.3 Dilatometry

Intuition.

Some liquid-phase reactions occur with a small but measurable change in total volume, because reactants and products have different molar volumes. A **dilatometer** — a reaction bulb fitted with a narrow graduated capillary tube, rather like a giant thermometer — converts this tiny volume change into a large, readable movement of the meniscus in the capillary.

PROPERTY **Principle of dilatometry**

If the volume of the reaction mixture changes linearly with the extent of reaction x, i.e. $V_t = V_0 + \alpha x$, then the height h of the liquid in the capillary (of uniform cross-section A) is a linear measure of x:

$$\frac{x_t}{x_\infty} = \frac{h_t - h_0}{h_\infty - h_0}.$$

The method has been classically applied to reactions such as the hydrolysis of esters and the inversion of sucrose, where the products occupy a different volume from the reactants.

The great advantage is simplicity: no reagent, no sampling, just a ruler reading against time. The limitations are equally clear: the volume change must be large enough to detect (typically $> 0.1\,\mathrm{cm}^3$), and temperature must be controlled to a fraction of a degree (e.g. $\pm 0.01\,^\circ\mathrm{C}$), because thermal expansion of the solvent would swamp the chemical volume change.

3.4 Gas Syringe and Pressure Measurement

Intuition.

When a reaction produces a gas, the gas itself is the most natural thing to measure. Think of adding dilute hydrochloric acid to marble chips (a familiar sight in every Cameroonian laboratory):

$$\mathrm{CaCO_3(s) + 2\,HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l) + CO_2(g).}$$

Every molecule of CO_2 released corresponds to one formula unit of $CaCO_3$ consumed. Two experimental arrangements are used.

(a) Gas syringe (constant pressure, variable volume).

The reaction flask is connected to a graduated gas syringe. As gas is evolved, the plunger is pushed out and the volume of gas is read at recorded times. The experiment is done at atmospheric pressure and room temperature. This method is ideal for reactions of moderate speed.

(b) Pressure measurement (constant volume, variable pressure).

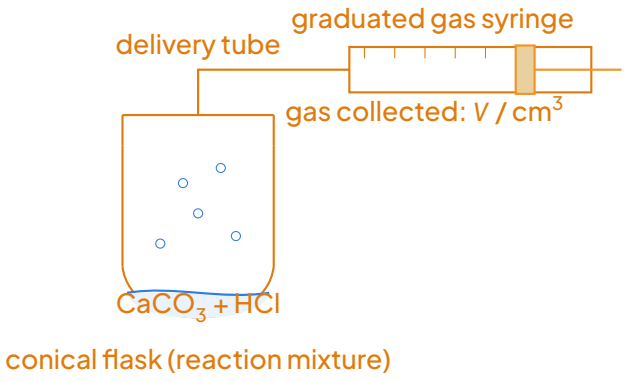
The reaction is carried out in a sealed vessel of fixed volume connected to a manometer or pressure sensor. As gas accumulates, the pressure rises, and the pressure–time trace replaces the volume–time trace. This method is more sensitive and can follow very fast gas-evolving reactions.

PROPERTY **From gas measurements to moles**

In both cases the amount of gas is obtained from the **ideal gas law**:

$$n = \frac{PV}{RT}, \qquad R = 8.314\ \text{J K}^{-1}\,\text{mol}^{-1}.$$

Gas syringe: P and T are constant, so $n \propto V$. Pressure method: V and T are constant, so $n \propto P$. Plotting n (or directly V or P) against time gives the rate curve; the gradient at any point is the rate of gas production, and the initial gradient gives the initial rate.



Gas syringe method: the CO_2 evolved pushes the plunger outwards; the volume is read on the graduated barrel at measured times. Plotting V (or $n = PV/RT$) against time gives the rate curve.

WORKED EXAMPLE 4 **Pressure method and the ideal gas law**

A 0.500 g sample of magnesium ribbon reacts with excess dilute hydrochloric acid in a sealed flask of volume 250 cm^3 at 25 °C:

$$\text{Mg(s)} + 2\,\text{HCl(aq)} \longrightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}.$$

(a) Calculate the final pressure rise due to the H_2 produced. (b) The pressure rises by

50 kPa after 40 s; find the average rate of H_2 production in mol s^{-1} over this interval. (M(Mg) = 24.3 g mol^{-1}; gas volume of the flask available: 250 cm^3.)

SOLUTION

(a) Moles of Mg: $n = 0.500/24.3 = 2.06\times10^{-2}$ mol, so $n(H_2) = 2.06\times10^{-2}$ mol (1:1 stoichiometry, acid in excess). Then

$$P = \frac{nRT}{V} = \frac{2.06\times10^{-2}\times8.314\times298}{250\times10^{-6}} = 2.04\times10^{5}\ \text{Pa} \approx 204\ \text{kPa}.$$

(b) A rise of 50 kPa corresponds to

$$n(H_2) = \frac{PV}{RT} = \frac{50\times10^{3}\times250\times10^{-6}}{8.314\times298} = 5.04\times10^{-3}\ \text{mol},$$

so the average rate is $5.04\times10^{-3}/40 = 1.3\times10^{-4}$ mol s^{-1}.

WARNING **Pitfalls of gas methods**

- Check for leaks: a poorly fitting bung or syringe gives low, erratic readings.
- The gas must be (nearly) insoluble in the reaction mixture; CO_2 dissolves appreciably in water, so early readings may underestimate the rate. Use a saturated solution or correct for solubility.
- Record temperature and atmospheric pressure — they are needed for $n = PV/RT$.
- The syringe plunger must move freely; friction delays the reading.
- Start timing at the moment of mixing, and mix quickly and thoroughly.

3.5 Choosing the Right Physical Method

Method	Quantity measured	Best suited for
Colorimetry	Absorbance $A = \varepsilon l c$	Reactions with one coloured species
Conductimetry	$\kappa = \sum \lambda_i c_i$	Reactions producing/consuming ions
Dilatometry	Liquid level in capillary	Liquid reactions with volume change
Gas syringe	Gas volume at constant P	Gas-evolving reactions (moderate speed)
Pressure sensor	Gas pressure at constant V	Gas-evolving reactions (fast or slow)

KEY POINTS **Key points of Section 3**

- Physical methods monitor a property proportional to concentration continuously and without disturbing the mixture.
- Colorimetry: $A = \varepsilon l c$; use a calibration line and a wavelength where only one species absorbs (λ_{max}).
- Conductimetry: $\kappa = \sum_i \lambda_i c_i$; large effects when H^+ or OH^- are involved; use AC current and thermostatting.
- Dilatometry: capillary rise measures extent of reaction via volume change; needs excellent temperature control (±0.01 °C).
- Gas methods: $n = PV/RT$; plot n, V or P against time and take gradients for rates.
- Whatever the method, the final step is identical: convert the measured quantity to concentration, plot c vs t, and measure the gradient — which prepares us for the rate laws of Section 4.

EXERCISE 3 **Choosing and exploiting a method**

For each reaction, state the most appropriate physical method and justify your choice:

1. $H_2O_2(aq) \longrightarrow H_2O(l) + \frac{1}{2} O_2(g)$ (catalysed by MnO_2);
2. the reaction between acidified $KMnO_4$ and $H_2C_2O_4$;
3. the hydrolysis of CH_3COOCH_3 by dilute NaOH.

SOLUTION **Exercise 1**

1. Gas syringe (or pressure sensor): O_2 gas is evolved and is insoluble enough; $n(O_2) = PV/RT$.
2. Colorimetry: MnO_4^- is intensely purple and the products are colourless; follow A at λ_{max} of MnO_4^- and use $A = \varepsilon l c$.
3. Conductimetry: the fast OH^- ions are replaced by slower CH_3COO^- ions, so κ decreases measurably; $[OH^-]_t \propto (\kappa_t - \kappa_\infty)$.

4 RATE LAW, ORDER OF REACTION, AND DETERMINATION METHODS

In the previous section we learned how to measure how fast a reaction goes: by titrating samples, by following colour, conductivity, volume or pressure. But a list of measurements is not yet a law of nature. The real power of kinetics appears when we compress all those measurements into a single mathematical sentence: the **rate law**. This section is the heart of the chapter. Once you can write rate = $k[A]^m[B]^n$ and determine k, m and n from experimental data, you can predict the rate of a reaction under any conditions — exactly what a chemical engineer designing a plant in Douala or Limbe needs.

4.1 The Rate Law and the Rate Constant

From observation to law.

Consider the decomposition of hydrogen peroxide, a reaction you can follow easily in the school laboratory with a gas syringe:

$$2\,H_2O_2(aq) \longrightarrow 2\,H_2O(l) + O_2(g)$$

If you double the concentration of H_2O_2 (keeping temperature and catalyst fixed), the initial rate doubles. If you triple it, the rate triples. The rate is directly proportional to $[H_2O_2]$. For other reactions, doubling a concentration may quadruple the rate, or may change nothing at all. Experiment — not the balanced equation — tells us how the rate depends on each concentration.

DEFINITION **Rate of reaction and rate law**

For a reaction aA + bB ⟶ products, the **rate of reaction** v is defined as

$$v = -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = \frac{1}{c}\frac{d[C]}{dt} = \dots$$

with units mol dm^{-3} s^{-1}. The **rate law** (or **rate equation**) is the experimentally determined relationship between this rate and the concentrations of the species that influence it:

$$v = k[A]^m[B]^n$$

where:

- k is the **rate constant** (or rate coefficient);
- m is the **partial order** of reaction with respect to A;
- n is the partial order with respect to B;
- $m + n$ is the **overall order** of the reaction.

The exponents m and n are found only by experiment; they are not the stoichiometric coefficients a and b.

WARNING **Orders are not coefficients!**

For $2\,N_2O_5(g) \longrightarrow 4\,NO_2(g) + O_2(g)$, experiment gives $v = k[N_2O_5]$, first order — the coefficient 2 does not appear as an exponent. For $NO_2(g) + CO(g) \longrightarrow NO(g) + CO_2(g)$, experiment gives $v = k[NO_2]^2$: the reaction is zero order in CO even though CO is a reactant. Never read orders from the balanced equation.

DEFINITION **The rate constant k**

The **rate constant** k is the proportionality factor in the rate law. Numerically, it equals the rate of reaction when all concentration terms equal 1 mol dm^{-3}. Its value:

- is constant for a given reaction at a given temperature;
- increases with temperature (its temperature dependence is described by the Arrhenius equation, Section 5);
- does not depend on concentrations;
- changes if a catalyst is used, because the catalyst provides a different reaction pathway.

Meaning of the orders.

- **Zero order** ($m = 0$): the rate is independent of [A]; $v = k$. Typical of reactions occurring on a saturated catalyst surface.
- **First order** ($m = 1$): doubling [A] doubles the rate.
- **Second order** ($m = 2$): doubling [A] multiplies the rate by four.

PROPERTY **Units of the rate constant**

Since the rate always has units of mol dm^{-3} s^{-1}, rearranging $k = v/[A]^m[B]^n$ gives the general rule:

$$\text{units of } k = (\text{mol dm}^{-3})^{1-\text{overall order}}\ \text{s}^{-1}$$

Overall order	Example rate law	Units of k
0	$v = k$	mol dm^{-3} s^{-1}
1	$v = k[A]$	s^{-1}
2	$v = k[A]^2$ or $k[A][B]$	dm^3 mol^{-1} s^{-1}
3	$v = k[A]^2[B]$	dm^6 mol^{-2} s^{-1}

A quick way to check a proposed rate law in the examination: work out the units of k — if they do not match the data, the law is wrong.

4.2 The Initial-Rates Method

The idea.

The cleanest way to find the orders is to measure the initial rate of the reaction (the gradient of the concentration–time curve at $t = 0$, Section 2) for several experiments in which the initial concentrations are changed one at a time. At $t = 0$ no products have accumulated, so no reverse or side reactions complicate the measurement.

METHOD **Finding orders by the initial-rates (ratio) method**

1. Tabulate the experiments: initial concentrations $[A]_0$, $[B]_0$ and measured initial rate v_0 for each run.
2. Choose two experiments where [B] is constant and only [A] changes. Write the ratio:

$$\frac{v_1}{v_2} = \left(\frac{[A]_1}{[A]_2}\right)^m$$

   and solve for m (if the concentration doubles and the rate quadruples, $2^m = 4$, so $m = 2$).
3. Repeat with a pair of experiments where [A] is constant to find n.
4. Write the complete rate law, then substitute the data of any one experiment to calculate k, with its units.

WORKED EXAMPLE 5 **Initial rates: the reaction between NO and H_2**

The reaction $2\,NO(g) + 2\,H_2(g) \longrightarrow N_2(g) + 2\,H_2O(g)$ was studied at 800 °C:

Expt	$[NO]_0$ / mol dm^{-3}	$[H_2]_0$ / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	0.10	0.10	1.2×10^{-3}
2	0.20	0.10	4.8×10^{-3}
3	0.20	0.20	9.6×10^{-3}

Determine the rate law, the overall order and the value of k.

SOLUTION

Order in NO: compare experiments 1 and 2 ($[H_2]$ constant). [NO] doubles (0.10 → 0.20) and the rate is multiplied by 4.8/1.2 = 4. Hence $2^m = 4$, so $m = 2$.
Order in H_2: compare experiments 2 and 3 ([NO] constant). $[H_2]$ doubles and the rate doubles (9.6/4.8 = 2). Hence $2^n = 2$, so $n = 1$.
Rate law: $v = k[NO]^2[H_2]$; overall order = 2 + 1 = 3.

Rate constant (using experiment 1):

$$k = \frac{v}{[NO]^2[H_2]} = \frac{1.2 \times 10^{-3}}{(0.10)^2(0.10)} = 1.2\ \text{dm}^6\ \text{mol}^{-2}\ \text{s}^{-1}$$

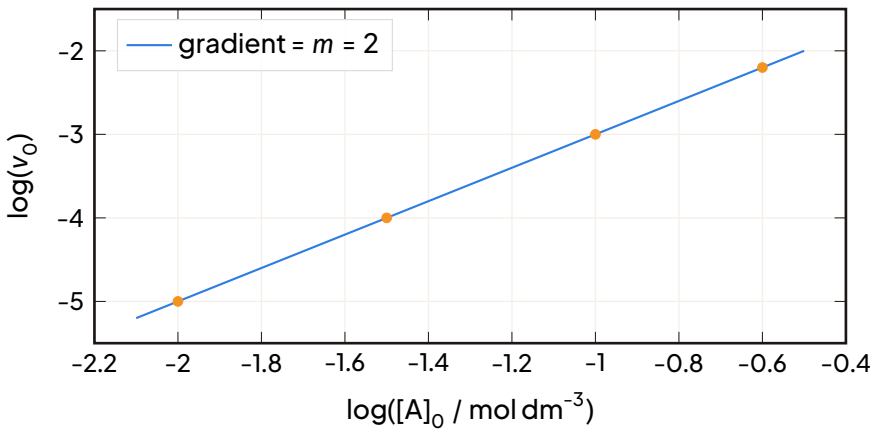
Check the units: $(\text{mol dm}^{-3})^{1-3}\text{s}^{-1} = \text{dm}^6\ \text{mol}^{-2}\ \text{s}^{-1}$. √

The log–log alternative.

When only one reactant influences the rate, $v = k[A]^m$ can be linearised by taking logarithms:

$$\log(v_0) = \log k + m \log[A]_0$$

A plot of log(initial rate) against $\log[A]_0$ is a straight line whose **gradient is the order** m and whose intercept is $\log k$. This is especially useful when the order is not an obvious integer.



Log–log plot of initial rate against initial concentration. The slope gives the order directly: here $m = 2$, and the intercept gives $\log k$.

4.3 Graphical Method: Integrated Rate Laws

Why integrate?

The initial-rates method needs several experiments. If instead we follow one reaction mixture over time (using any method of Section 3), we can find the order from the shape of the concentration–time data, using the integrated forms of the rate laws. For reactions with more than one reactant, we use the **isolation method** (or **pseudo-order method**): one reactant is in large excess so its concentration stays effectively constant, and the rate law reduces to $v = k'[A]^m$ where $k' = k[B]_0^n$ is a pseudo-rate constant.

PROPERTY **Integrated rate laws (statements and linear plots)**

For a reactant A of concentration $[A]_t$ at time t and $[A]_0$ at $t = 0$:

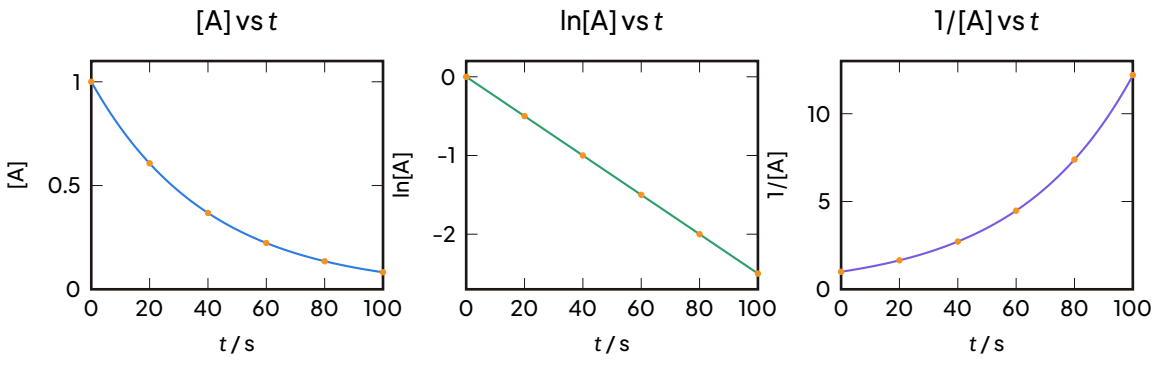
- Zero order ($-d[A]/dt = k$): $[A]_t = [A]_0 - kt$. A plot of [A] vs t is linear, gradient $= -k$.
- First order ($-d[A]/dt = k[A]$): $\ln[A]_t = \ln[A]_0 - kt$. A plot of ln[A] vs t is linear, gradient $= -k$.
- Second order ($-d[A]/dt = k[A]^2$): $\frac{1}{[A]_t} = \frac{1}{[A]_0} + kt$. A plot of 1/[A] vs t is linear, gradient $= +k$.

The first-order derivation is examinable: separating variables,

$$\int_{[A]_0}^{[A]_t} \frac{d[A]}{[A]} = -k \int_0^t dt \quad \Longrightarrow \quad \ln[A]_t - \ln[A]_0 = -kt.$$

METHOD **Identifying the order graphically**

1. From the experimental data, compute [A], ln[A] and 1/[A] at each time.
2. Plot the three graphs: [A] vs t, ln[A] vs t, 1/[A] vs t.
3. Only one of the three is a straight line: that plot identifies the order.
4. Read the gradient of the straight line to obtain k (with the correct units).



The same first-order data set tested three ways. Only the ln[A] vs t plot is linear (gradient $= -k$): the reaction is first order. The curved [A] vs t and 1/[A] vs t plots confirm it is neither zero nor second order.

WORKED EXAMPLE 6 **Graphical test for the hydrolysis of an ester**

The hydrolysis of methyl ethanoate in excess dilute acid was followed by titration (Section 2). The concentration of ester remaining was:

t / min	0	20	40	60	80
[ester] / mol dm^{-3}	0.500	0.389	0.303	0.236	0.184
ln[ester]	-0.693	-0.944	-1.194	-1.444	-1.693

Show the reaction is first order in ester and find k.

SOLUTION

The values of ln[ester] fall by almost exactly 0.250 every 20 minutes: the plot of ln[ester] against t is a straight line, so the reaction is first order in ester (the acid, in large excess, does not appear — its concentration is effectively constant, giving a pseudo-first-order rate constant k').
Gradient of the line:

$$-k' = \frac{-1.693 - (-0.693)}{80 - 0} = \frac{-1.000}{80} = -0.0125\ \text{min}^{-1}$$

Hence $k' = 0.0125\ \text{min}^{-1} = 2.08e-4\ \text{s}^{-1}$. (First order, so the unit is s^{-1}.)

4.4 Half-Life of a First-Order Reaction

DEFINITION **Half-life**

The **half-life** $t_{1/2}$ of a reaction is the time taken for the concentration of a reactant to fall to half of its initial value.

For a first-order reaction, set $[A]_t = [A]_0/2$ in the integrated law:

$$\ln\left(\frac{[A]_0}{[A]_0/2}\right) = kt_{1/2} \quad \Longrightarrow \quad \ln 2 = kt_{1/2} \quad \Longrightarrow \quad \boxed{t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k}}$$

PROPERTY **First-order half-life is constant**

For a first-order reaction, $t_{1/2} = \ln 2/k$ depends only on k, not on the initial concentration. The concentration therefore falls in the pattern $[A]_0 \rightarrow [A]_0/2 \rightarrow [A]_0/4 \rightarrow [A]_0/8$ after 1, 2, 3 half-lives. This constant half-life is a quick experimental test for first order: if the time to halve is the same wherever you start on the curve, the reaction is first order.
For comparison:

- Zero order: $t_{1/2} = \dfrac{[\mathrm{A}]_0}{2k}$ (decreases as reaction proceeds).
- Second order: $t_{1/2} = \dfrac{1}{k[\mathrm{A}]_0}$ (increases as reaction proceeds).

EXEMPLE **Half-life of N_2O_5 decomposition**

For $2\,N_2O_5(g) \longrightarrow 4\,NO_2(g) + O_2(g)$, $k = 6.2e-4\ s^{-1}$ at 45 °C. Then

$$t_{1/2} = \frac{0.693}{6.2 \times 10^{-4}} \approx 1.1e3\ \mathrm{s} \approx 19\ \mathrm{min}.$$

Whatever the starting pressure of N_2O_5, half of it disappears every 19 minutes at this temperature.

WARNING **Fractional and negative orders exist**

Do not force the data to give order 0, 1 or 2. Many real reactions have fractional orders (e.g. $v = k[CH_3CHO]^{3/2}$ for the thermal decomposition of ethanal) or even negative orders, because they proceed through multi-step mechanisms. If your ratio calculation gives $2^m = 2.8$, take logarithms: $m = \log 2.8 / \log 2 \approx 1.5$. Also remember: the order is defined only with respect to concentration; k itself changes with temperature.

EXERCISE 4 **Practice: orders and half-life**

1. For the reaction A + B ⟶ C, the rate doubles when [A] doubles (at fixed [B]) and is unchanged when [B] doubles. Write the rate law, state the overall order, and give the units of k.
2. A first-order reaction has $t_{1/2} = 346$ s. Calculate k, and the time needed for the concentration to fall to one eighth of its initial value.
3. Using the data of the worked example on ester hydrolysis, verify that the half-life is constant by computing the time to go from 0.500 to 0.250 and from 0.400 to 0.200 mol dm^{-3}.

SOLUTION **Exercise 4.4**

1. Rate $v = k[A]$; first order in A, zero order in B, overall order 1; units of k: s^{-1}.
2. $k = 0.693/346 = 2.00e-3\ s^{-1}$. One eighth means three half-lives: $t = 3 \times 346 = 1038$ s.
3. From the integrated law, $t_{1/2} = \ln 2/k' = 0.693/0.0125 = 55.4$ min.
   - From 0.500 to 0.250: $\ln(0.250) = -1.386$. Interpolating between $t = 40$ ($\ln = -1.194$) and $t = 60$ ($\ln = -1.444$): $t \approx 40 + \frac{1.386-1.194}{1.444-1.194} \times 20 = 55.4$ min.
   - From 0.400 to 0.200: $\ln(0.400) = -0.916$, $\ln(0.200) = -1.609$. Interpolating: $t_{0.400} \approx 17.8$ min, $t_{0.200} \approx 73.2$ min. Difference = 55.4 min.

   Equal half-lives confirm first order. √

KEY POINTS **Key points of Section 4**

- Rate law: $v = k[A]^m[B]^n$; orders m, n come from experiment only, never from stoichiometric coefficients.
- Units of k: $(\text{mol dm}^{-3})^{1-\text{overall order}}\text{s}^{-1}$.
- Initial-rates method: vary one concentration at a time and use $v_1/v_2 = ([A]_1/[A]_2)^m$; a log–log plot gives the order as its gradient.
- Graphical method: the linear plot among [A], ln[A], 1/[A] vs t identifies zero, first or second order; the gradient gives k. For multi-reactant systems, use the isolation (pseudo-order) method.
- First-order half-life: $t_{1/2} = \ln 2/k$, independent of initial concentration — a diagnostic test for first order.
- Orders may be fractional or zero; let the data speak.

With the rate law firmly in hand, one deep question remains: why does the rate constant depend so strongly on temperature, and what actually happens when two molecules meet? That is the subject of the next section: collision theory, the Maxwell–Boltzmann distribution, the Arrhenius equation and transition-state theory.

5 THEORETICAL MODELS: COLLISION THEORY, ARRHENIUS EQUATION, AND TRANSITION-STATE THEORY

In the previous section we learned how to express the rate of a reaction as a rate law, and how to determine the order of reaction and the rate constant k from experimental data. We now ask a deeper question: why does the rate constant have the value it does, and how does it change with temperature? The answer lies in the microscopic picture of reacting particles. Three com-

plementary models — Collision Theory, the Arrhenius Equation, and Transition-State Theory — provide that picture. They are the theoretical backbone of chemical kinetics and are examined in detail in the GCE Advanced Level syllabus (Lessons 98–99).

5.1 Collision Theory and Effective Collisions

Collision theory starts from a simple, intuitive idea: for a reaction to occur, reactant particles must **collide**. However, not every collision leads to products. Two conditions must be satisfied simultaneously:

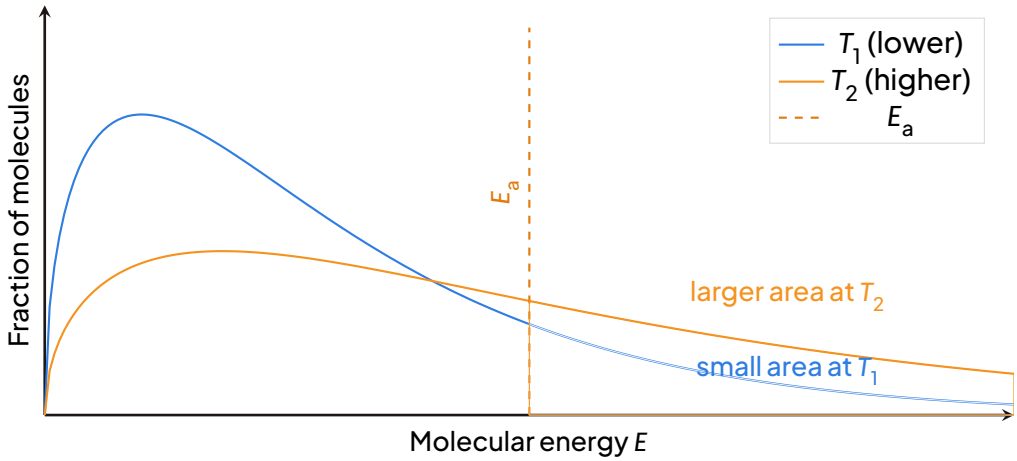
1. **Energy requirement:** the colliding particles must together possess kinetic energy at least equal to the **activation energy** E_a. This is the minimum energy needed to start breaking the existing bonds so that new ones can form.
2. **Orientation requirement:** the colliding particles must be oriented in a way that allows the relevant atoms to interact. For example, in the reaction $NO + O_3 \longrightarrow NO_2 + O_2$, the nitrogen atom of NO must strike a terminal oxygen atom of O_3; a collision in which the oxygen end of NO hits O_3 is ineffective.

Only collisions fulfilling both criteria are called **effective collisions**. The rate of reaction is therefore proportional to the frequency of effective collisions.

DEFINITION **Activation Energy E_a**

The minimum energy that reacting particles must possess for a collision between them to result in a chemical reaction. It is the energy barrier separating reactants from products.

The fraction of molecules that have energy greater than or equal to E_a at a given temperature T is obtained from the **Maxwell–Boltzmann distribution** of molecular energies. As temperature rises, the distribution broadens and its maximum shifts to higher energies, while the area under the curve (the total number of molecules) stays constant. The fraction of molecules with energy at least E_a therefore increases sharply with temperature.



Maxwell–Boltzmann energy distributions at two temperatures. The shaded area to the right of E_a represents the fraction of molecules with sufficient energy to react. Raising the temperature from T_1 to T_2 increases this fraction considerably, which is why rates increase so rapidly with temperature.

For a bimolecular gas-phase reaction between molecules A and B, the collision frequency Z (number of collisions per unit volume per unit time) is proportional to the product of the concentrations [A][B] and to $\sqrt{T}$. Only a fraction $f = e^{-E_a/RT}$ of these collisions have enough energy, and only a fraction p (the **steric** or **orientation factor**, usually $0 < p \leq 1$) have the correct orientation. The rate constant predicted by simple collision theory is therefore

$$k = p\,Z_0\,e^{-E_a/RT}$$

where Z_0 is the collision frequency at unit concentrations. This already resembles the Arrhenius form, with the pre-exponential factor $A \approx pZ_0$ (with a weak additional dependence on $\sqrt{T}$).

WARNING **The Steric Factor**

For many reactions $p < 1$ because only a fraction of collisions have the correct geometry. Do not state, however, that p is always less than 1: for some reactions (for example "harpoon" reactions between alkali metals and halogens, where an electron is transferred at long range) the effective collision cross-section is larger than the geometric one and p can exceed 1.

5.2 The Arrhenius Equation

Svante Arrhenius (1889) proposed an empirical relationship that fits the temperature dependence of the vast majority of rate constants:

$$k = A\,e^{-E_a/RT}$$

where

- k = rate constant (units depend on the overall order),

- A = **pre-exponential factor** (or frequency factor), with the same units as k,
- E_a = activation energy (J mol^{-1} or kJ mol^{-1}),
- R = gas constant (8.314 J mol^{-1} K^{-1}),
- T = absolute temperature (K).

Taking natural logarithms gives the linear form:

$$\ln k = \ln A - \frac{E_a}{R}\frac{1}{T}$$

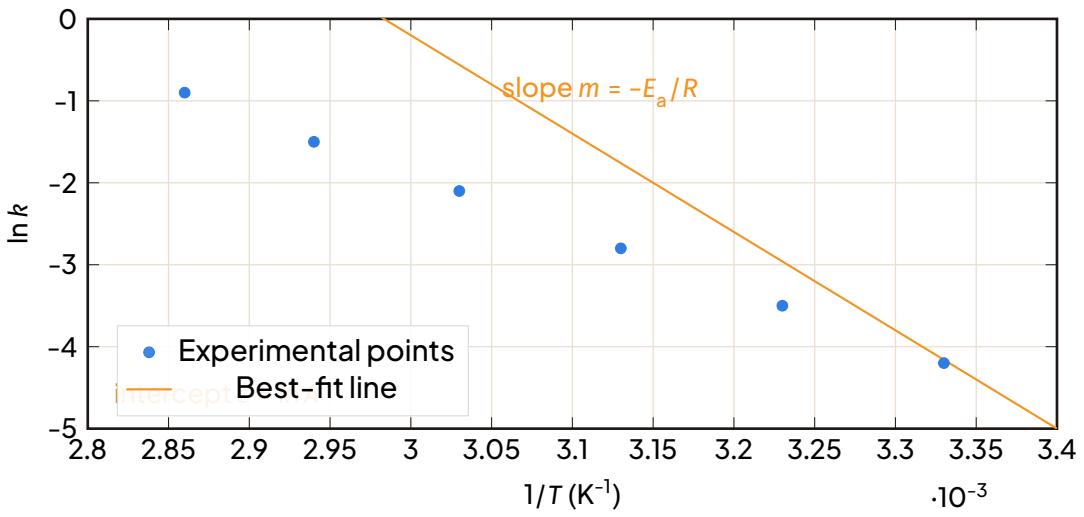
A plot of $\ln k$ against $1/T$ (an **Arrhenius plot**) is a straight line of slope $-E_a/R$ and intercept $\ln A$. This is the standard experimental method for measuring activation energies.

PROPERTY **Meaning of the Pre-exponential Factor A**

A is related to the frequency of collisions and to the probability that a collision has the correct orientation. In collision theory $A \approx pZ_0$; in transition-state theory $A = (k_B T/h)\, e^{\Delta S^{\ddagger}/R}$. A depends weakly on temperature, but over modest temperature ranges it is treated as constant. (The identification with the Eyring parameters is examinable as a comparison, not as a formal proof.)

METHOD **Determining E_a and A from an Arrhenius Plot**

1. Measure the rate constant k at several different temperatures (at least three, preferably five or six).
2. Convert each temperature to kelvin and compute $1/T$ (K^{-1}) and $\ln k$.
3. Plot $\ln k$ (vertical axis) against $1/T$ (horizontal axis).
4. Draw the best-fit straight line and determine its slope m and intercept c.
5. Calculate $E_a = -mR$ (in J mol^{-1}, then divide by 1000 for kJ mol^{-1}) and $A = e^c$ (same units as k).



Typical Arrhenius plot. The straight line confirms Arrhenius behaviour. From the slope $m = -1.20 \times 10^4$ K we obtain $E_a = -mR = (1.20 \times 10^4)(8.314) = 99.8$ kJ mol^{-1}.

WARNING **Apparent Activation Energy**

For complex (multi-step) reactions the observed rate constant may be a combination of several elementary rate constants. The E_a obtained from an Arrhenius plot is then an apparent activation energy, which can vary with temperature if the rate-determining step changes. Always check the linearity of the Arrhenius plot; curvature signals a change of mechanism.

WORKED EXAMPLE 7 **Determining E_a from Two Temperatures**

The rate constant for the decomposition of N_2O_5 in CCl_4 is $k_1 = 3.4 \times 10^{-5}$ s^{-1} at $T_1 = 298$ K and $k_2 = 1.2 \times 10^{-3}$ s^{-1} at $T_2 = 318$ K. Assuming Arrhenius behaviour, calculate the activation energy E_a.

SOLUTION

Solution: Dividing the Arrhenius equation at T_2 by that at T_1 eliminates A:

$$\ln\frac{k_2}{k_1} = -\frac{E_a}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$\ln\frac{1.2 \times 10^{-3}}{3.4 \times 10^{-5}} = \ln(35.3) = 3.56$$

$$\frac{1}{T_2} - \frac{1}{T_1} = \frac{1}{318} - \frac{1}{298} = (3.145 - 3.356) \times 10^{-3} = -2.11 \times 10^{-4}\ \text{K}^{-1}$$

$$E_a = -\frac{R\ln(k_2/k_1)}{1/T_2 - 1/T_1} = -\frac{8.314 \times 3.56}{-2.11 \times 10^{-4}} = 1.40 \times 10^5\ \text{J mol}^{-1} = \mathbf{140}\ \text{kJ mol}^{-1}$$

5.3 Transition-State Theory (Activated Complex Theory)

Transition-State Theory (TST), developed by Eyring, Evans and Polanyi in the 1930s, gives a more detailed microscopic picture. It postulates that reactants pass through a high-energy, transient arrangement of atoms called the **activated complex** or **transition state** (denoted ‡), which is assumed to be in quasi-equilibrium with the reactants:

$$A + B \rightleftharpoons AB^{\ddagger} \rightarrow \text{Products}$$

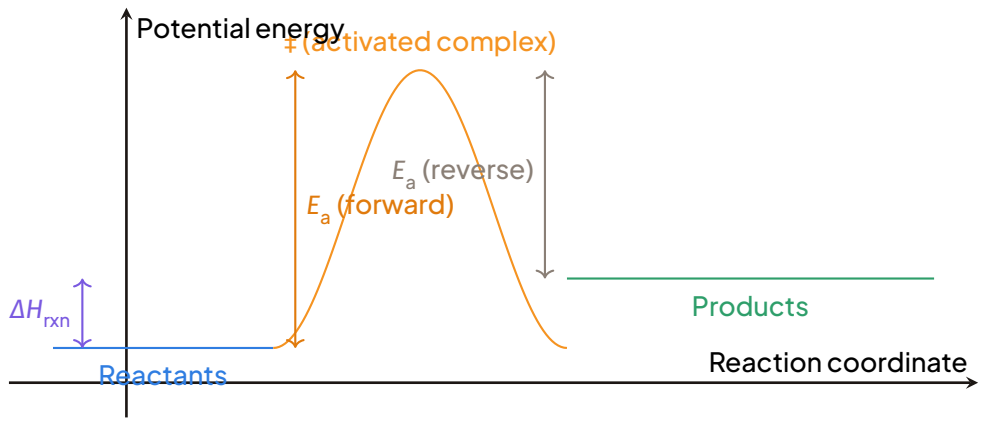
The rate of reaction is the rate at which activated complexes pass over the energy barrier to products. Assuming a universal frequency of passage $\nu = k_B T/h$ (where k_B is Boltzmann's constant and h is Planck's constant), one obtains the **Eyring equation**:

$$k = \frac{k_B T}{h} e^{-\Delta G^{\ddagger}/RT} = \frac{k_B T}{h} e^{\Delta S^{\ddagger}/R} e^{-\Delta H^{\ddagger}/RT}$$

where $\Delta G^{\ddagger} = \Delta H^{\ddagger} - T\Delta S^{\ddagger}$ is the Gibbs free energy of activation, $\Delta H^{\ddagger}$ the enthalpy of activation and $\Delta S^{\ddagger}$ the entropy of activation.

DEFINITION **Activated Complex (‡)**

A transient, high-energy arrangement of atoms corresponding to the maximum of the potential energy profile along the reaction coordinate. It has a very short lifetime (of the order of 10^{-13} s) and must not be confused with a stable reaction intermediate, which sits in an energy minimum and can in principle be detected.



Potential energy profile for an exothermic reaction. The activated complex (‡) sits at the energy maximum. For an exothermic reaction $E_a(\text{forward}) < E_a(\text{reverse})$, and $\Delta H_{rxn} = E_a(\text{forward}) - E_a(\text{reverse}) < 0$.

Comparing the Eyring equation with the Arrhenius equation,

$$k = \underbrace{\frac{k_B T}{h} e^{\Delta S^\ddagger / R}}_{A} \; e^{-\Delta H^\ddagger / RT}$$

we identify, for a gas-phase bimolecular reaction,

$$E_a = \Delta H^\ddagger + RT \qquad \text{and} \qquad A = \frac{k_B T}{h} e^{\Delta S^\ddagger / R}$$

The entropy of activation $\Delta S^\ddagger$ measures the change in order on forming the activated complex: bringing two separate molecules together into one complex usually gives $\Delta S^\ddagger < 0$, which lowers A and slows the reaction — a quantitative explanation of the steric factor p of collision theory.

5.4 Comparison of Collision Theory and Transition-State Theory

Both theories explain the temperature dependence of reaction rates, but they differ in scope and detail.

Aspect	Collision Theory	Transition-State Theory
Basic idea	Reactants must collide with sufficient energy and correct orientation.	Reactants form an activated complex in quasi-equilibrium; passage of ‡ over the barrier gives products.
Key parameters	E_a, steric factor p, collision frequency Z.	$\Delta H^\ddagger, \Delta S^\ddagger, \Delta G^\ddagger$.
Rate constant	$k = pZ_0 e^{-E_a/RT}$ (approx.)	$k = \frac{k_B T}{h} e^{-\Delta G^\ddagger / RT}$
Entropy effects	Not explicit; hidden in the empirical factor p.	Explicit via $\Delta S^\ddagger$; explains why some reactions with low E_a are still slow.
Applicability	Mainly gas-phase bimolecular reactions.	Gas phase, solution and surfaces; unimolecular and bimolecular.
Link with Arrhenius	$A \approx pZ_0$ (weak T dependence).	$A = \frac{k_B T}{h} e^{\Delta S^\ddagger / R}$; $E_a = \Delta H^\ddagger + RT$.

KEY POINTS **Key Points for the Examination**

- Collision theory: an effective collision requires both $E \geq E_a$ and correct orientation. The Maxwell–Boltzmann distribution shows how the fraction of molecules with $E \geq E_a$ increases with temperature.
- Arrhenius equation: $k = Ae^{-E_a/RT}$; linear form $\ln k = \ln A - E_a/(RT)$. Plot $\ln k$ against $1/T$: slope = $-E_a/R$, intercept = $\ln A$.

- Two-temperature form: $\ln(k_2/k_1) = -\frac{E_a}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$.
- Transition-state theory: activated complex ‡ in quasi-equilibrium with reactants; Eyring equation $k = (k_B T/h)\, e^{-\Delta G^{\ddagger}/RT}$.
- $\Delta G^{\ddagger} = \Delta H^{\ddagger} - T\Delta S^{\ddagger}$; for gas-phase reactions $E_a = \Delta H^{\ddagger} + RT$.
- TST includes the entropy of activation and so explains steric effects quantitatively; collision theory lumps them into the empirical factor p.
- The apparent E_a from an Arrhenius plot may not be constant for complex mechanisms.

WORKED EXAMPLE 8 **From Arrhenius to Eyring Parameters**

For a certain gas-phase reaction the Arrhenius parameters are $A = 2.5 \times 10^{13}$ dm^3 mol^{-1} s^{-1} and $E_a = 75.0$ kJ mol^{-1} at $T = 298$ K. Calculate $\Delta H^{\ddagger}$, $\Delta S^{\ddagger}$ and $\Delta G^{\ddagger}$. (Take $k_B = 1.381 \times 10^{-23}$ J K^{-1}, $h = 6.626 \times 10^{-34}$ J s.)

SOLUTION

Solution:

$$\Delta H^{\ddagger} = E_a - RT$$
$$= 75.0\ \text{kJ mol}^{-1} - (8.314 \times 10^{-3}\ \text{kJ mol}^{-1}\ \text{K}^{-1} \times 298\ \text{K})$$
$$= 75.0 - 2.48 = \mathbf{72.5}\ \text{kJ mol}^{-1}$$

$$A = \frac{k_B T}{h}\, e^{\Delta S^{\ddagger}/R} \quad \Rightarrow \quad \Delta S^{\ddagger} = R \ln\left(\frac{Ah}{k_B T}\right)$$

$$\frac{Ah}{k_B T} = \frac{(2.5 \times 10^{13})(6.626 \times 10^{-34})}{(1.381 \times 10^{-23})(298)} = \frac{1.657 \times 10^{-20}}{4.115 \times 10^{-21}} = 4.03$$

$$\Delta S^{\ddagger} = 8.314 \times \ln(4.03) = 8.314 \times 1.39 = +\mathbf{11.6}\ \text{J mol}^{-1}\ \text{K}^{-1}$$

$$\Delta G^{\ddagger} = \Delta H^{\ddagger} - T\Delta S^{\ddagger}$$
$$= 72.5\ \text{kJ mol}^{-1} - (298\ \text{K})(11.6 \times 10^{-3}\ \text{kJ mol}^{-1}\ \text{K}^{-1})$$
$$= 72.5 - 3.46 = \mathbf{69.0}\ \text{kJ mol}^{-1}$$

Remark: strictly, A and $k_B T/h$ have different units for a bimolecular reaction, so the ratio $Ah/(k_B T)$ implicitly uses the standard-state concentration (1 mol dm^{-3}); this convention is assumed in the GCE treatment.

EXERCISE 5 **Practice: Arrhenius Plot and TST**

The rate constant for the gas-phase reaction $2\,HI(g) \longrightarrow H_2(g) + I_2(g)$ was measured at three temperatures:

T / K	k / dm^3 mol^{-1} s^{-1}
500	1.2×10^{-3}
550	4.8×10^{-3}
600	1.6×10^{-2}

1. Construct an Arrhenius plot and determine E_a and A.
2. Using the values obtained, calculate $\Delta H^{\ddagger}$, $\Delta S^{\ddagger}$ and $\Delta G^{\ddagger}$ at 550 K.
3. Comment on the sign of $\Delta S^{\ddagger}$ in relation to the molecularity of the reaction.

SOLUTION **Exercise 5.1**

1. Compute $1/T$ and $\ln k$:

$1/T$ (K^{-1})	$\ln k$
2.000×10^{-3}	−6.73
1.818×10^{-3}	−5.34
1.667×10^{-3}	−4.14

The points are collinear. Using the extreme points:

$$m = \frac{-4.14 - (-6.73)}{1.667 \times 10^{-3} - 2.000 \times 10^{-3}} = \frac{2.59}{-3.33 \times 10^{-4}} = -7.77 \times 10^{3}\ \text{K}$$

$$E_a = -mR = (7.77 \times 10^{3})(8.314) = 6.46 \times 10^{4}\ \text{J mol}^{-1} = \mathbf{64.6}\ \text{kJ mol}^{-1}$$

Intercept: $c = \ln k - m(1/T) = -6.73 + (7.77 \times 10^{3})(2.000 \times 10^{-3}) = 8.81$, so

$$A = e^{8.81} = \mathbf{6.7 \times 10^{3}}\ \text{dm}^3\ \text{mol}^{-1}\ \text{s}^{-1}$$

2. At $T = 550$ K:

$$\Delta H^{\ddagger} = E_a - RT = 64.6 - (8.314 \times 10^{-3} \times 550) = 64.6 - 4.57 = \mathbf{60.0}\ \text{kJ mol}^{-1}$$

$$\Delta S^{\ddagger} = R \ln\left(\frac{Ah}{k_B T}\right) = 8.314 \ln\left(\frac{(6.7 \times 10^{3})(6.626 \times 10^{-34})}{(1.381 \times 10^{-23})(550)}\right)$$

$$= 8.314 \ln(5.8 \times 10^{-10}) = 8.314 \times (-21.3) = \mathbf{-177}\ \text{J mol}^{-1}\ \text{K}^{-1}$$

$$\Delta G^{\ddagger} = \Delta H^{\ddagger} - T\Delta S^{\ddagger} = 60.0 - (550)(-0.177) = 60.0 + 97.2 = \mathbf{157}\ \text{kJ mol}^{-1}$$

3. The reaction is bimolecular: two HI molecules must come together to form a single activated complex. This association reduces the freedom of motion (translational degrees of freedom) of the particles, so the system becomes more ordered and $\Delta S^{\ddagger}$ is expected to be negative — exactly what is found. The large negative $\Delta S^{\ddagger}$ makes $\Delta G^{\ddagger}$ much larger than $\Delta H^{\ddagger}$ and corresponds to a small steric factor p in collision theory: only a small fraction of sufficiently energetic collisions have the correct orientation.

DID YOU KNOW? **Did You Know? — The Arrhenius Equation in Everyday Life**

The Arrhenius equation explains why food spoils faster at room temperature than in a refrigerator. Taking a typical activation energy of about 80 kJ mol^{-1} for the chemical and microbial processes of spoilage, lowering the temperature from 25 °C (298 K) to 4 °C (277 K) changes the rate constant by the factor

$$\frac{k_{277}}{k_{298}} = \exp\left[-\frac{E_a}{R}\left(\frac{1}{277} - \frac{1}{298}\right)\right] = \exp(-2.45) \approx 0.086$$

i.e. spoilage becomes roughly 12 times slower. This is why cold storage is so valuable in Cameroon's tropical climate for preserving tomatoes, plantains, fish and meat between the market and the kitchen.

WARNING **Common Examination Traps**

- Forgetting to convert E_a to J mol^{-1} when using $R = 8.314$ J mol^{-1} K^{-1}.
- Using $\log_{10}$ instead of ln in the Arrhenius plot; with $\log_{10}$ the slope is $-E_a/(2.303R)$.
- Confusing E_a (Arrhenius) with $\Delta H^{\ddagger}$ (Eyring); remember $E_a = \Delta H^{\ddagger} + RT$ for gas-phase reactions.
- Confusing the activated complex (an energy maximum, undetectable) with a reaction intermediate (an energy minimum, detectable).
- Stating that the steric factor p is always less than 1; long-range attractions can make $p > 1$.
- Claiming that a catalyst lowers E_a "by giving molecules more energy": a catalyst provides an alternative pathway of lower activation energy; the temperature, and hence the molecular energies, are unchanged.

6 CATALYSIS: HOMOGENEOUS AND HETEROGENEOUS

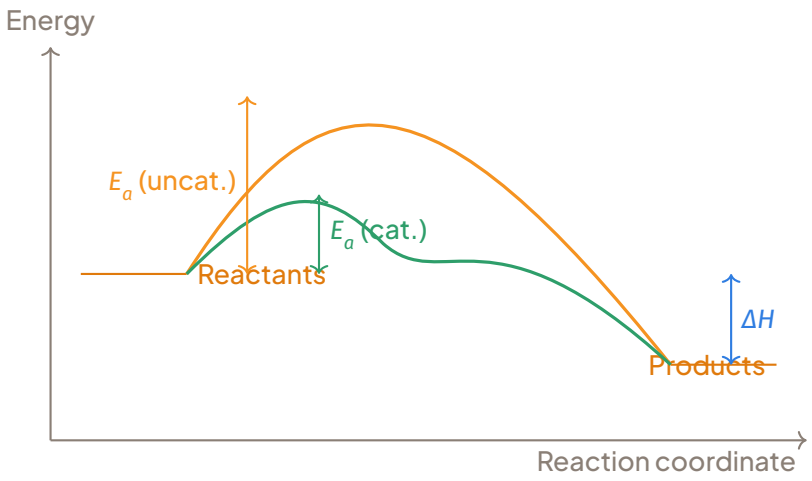
In the previous section we saw that the rate of a reaction is controlled by its **activation energy** E_a: the Arrhenius equation $k = Ae^{-E_a/RT}$ tells us that even a modest decrease in E_a produces an enormous increase in the rate constant k, because E_a appears in an exponential. This raises a practical question of immense industrial importance: can we lower the activation energy of a reaction without changing the reactants or the temperature? The answer is yes — this is precisely what a **catalyst** does. From the manufacture of ammonia that feeds fertiliser factories supplying Cameroonian agriculture, to the catalytic converters fitted in the exhaust pipes of taxis in Douala, catalysis is one of the most economically important applications of chemical kinetics.

6.1 What is a Catalyst?

DEFINITION **Catalyst**

A **catalyst** is a substance that increases the rate of a chemical reaction without itself being permanently consumed in the reaction. It is recovered, chemically unchanged, at the end of the reaction, although it may take part in intermediate steps.

The key to understanding catalysis lies in the energy profile. A catalyst does not simply "push" the reactants harder; it provides an **alternative reaction pathway** (a different mechanism, usually with several steps) whose highest energy barrier is **lower** than the activation energy E_a of the uncatalysed pathway. Since a much larger fraction of molecules now possesses enough energy to cross the barrier (recall the Maxwell–Boltzmann distribution), the rate increases dramatically.



Energy profile for an exothermic reaction. The catalysed pathway (green) proceeds through an intermediate (the dip between the two humps) with a lower activation energy than the uncatalysed pathway (orange). Note that ΔH is unchanged.

PROPERTY **What a catalyst does and does not do**

- A catalyst lowers the activation energy by providing an alternative mechanism, so the rate constant increases: $k = Ae^{-E_a/RT}$ grows as E_a falls.
- A catalyst does not change the enthalpy change ΔH, the free energy change ΔG, or the equilibrium constant K of the reaction.
- Because it lowers the barrier in both directions, a catalyst accelerates the forward and reverse reactions equally. It therefore helps the system to reach equilibrium faster, but the position of equilibrium is unchanged.
- Only a very small amount of catalyst is needed, since it is regenerated during the catalytic cycle.

(The thermodynamic statement — that catalysis cannot alter K because K depends only on ΔG° — is examinable as a short-answer justification.)

WARNING **A catalyst never makes an impossible reaction happen**

A common examination error is to write that a catalyst "makes a reaction occur". If a reaction is thermodynamically impossible ($\Delta G > 0$), no catalyst can make it go. A catalyst only speeds up a reaction that is already feasible but kinetically slow. Likewise, a catalyst does not appear in the overall balanced equation — if you find it as a reactant on the left and never regenerated, it is not a catalyst but a reagent. Finally, do not confuse an

intermediate (a real, sometimes isolable species sitting in an energy valley between two humps) with the transition state (the unstable arrangement at the top of a hump).

Catalysts are classified into two great families according to the phase they occupy relative to the reactants.

DEFINITION **Homogeneous and heterogeneous catalysts**

A **homogeneous catalyst** is in the same phase as the reactants (typically all in aqueous solution or all gases). A **heterogeneous catalyst** is in a different phase from the reactants — most often a solid catalyst in contact with gaseous or liquid reactants.

6.2 Homogeneous Catalysis

In homogeneous catalysis, the catalyst mixes freely with the reactants at the molecular level, so every catalyst molecule is accessible. The mechanism proceeds through the formation of a reactive **intermediate**: the catalyst is consumed in one step and regenerated in a later one.

Example 1: acid-catalysed ester hydrolysis. The hydrolysis of an ester such as ethyl ethanoate is slow with pure water but rapid in the presence of dilute HCl or H_2SO_4, which supply H^+ ions:

$$CH_3COOC_2H_5(aq) + H_2O(l) \xrightarrow{H^+} CH_3COOH(aq) + C_2H_5OH(aq)$$

The H^+ ion first protonates the carbonyl oxygen of the ester, making the carbonyl carbon much more susceptible to nucleophilic attack by water. After a series of proton transfers, the products are released and H^+ is regenerated. Each step of this catalysed route has a lower activation energy than the direct attack of water on the neutral ester.

Example 2: the persulfate–iodide reaction. The oxidation of iodide by peroxodisulfate,

$$S_2O_8^{2-}(aq) + 2\,I^-(aq) \longrightarrow 2\,SO_4^{2-}(aq) + I_2(aq),$$

is slow because it brings two negative ions together (strong electrostatic repulsion). It is catalysed by Fe^{2+} or Fe^{3+} ions, which provide a two-step route in which each step pairs ions of opposite charge:

$$S_2O_8^{2-} + 2\,Fe^{2+} \longrightarrow 2\,SO_4^{2-} + 2\,Fe^{3+}$$
$$2\,Fe^{3+} + 2\,I^- \longrightarrow 2\,Fe^{2+} + I_2$$

The iron ions are consumed in the first step and regenerated in the second — the signature of a true catalyst. Adding the two steps regenerates the overall equation, with no iron species appearing in it.

Example 3: enzyme catalysis. Enzymes are biological catalysts of extraordinary efficiency and specificity. Amylase in saliva hydrolyses starch; catalase decomposes hydrogen peroxide into

water and oxygen almost instantaneously. Enzymes work by binding the reactant (the substrate) at a specifically shaped active site, holding it in the ideal geometry for reaction — a beautiful molecular illustration of transition-state stabilisation.

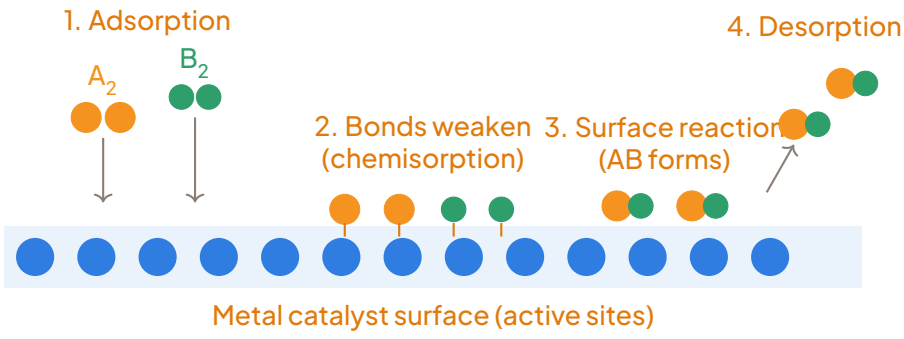
DID YOU KNOW? **Catalysis and life**

A single molecule of catalase can decompose millions of hydrogen peroxide molecules per second. Without enzymes, the biochemical reactions that keep us alive would be far too slow at body temperature — life itself is, in a sense, a triumph of catalysis. The fermentation of palm wine and of cassava dough for garri, familiar processes in Cameroon, are also driven by enzyme catalysis.

6.3 Heterogeneous Catalysis

In heterogeneous catalysis the reaction occurs at the **surface** of the catalyst. The mechanism is usually described in four stages:

1. **Diffusion** of reactant molecules to the surface.
2. **Adsorption**: reactant molecules bind to **active sites** on the surface. Two types are distinguished: physisorption (weak van der Waals attraction, easily reversed) and chemisorption (formation of genuine chemical bonds with surface atoms — the step that activates the molecule, often by weakening or breaking its internal bonds).
3. **Surface reaction**: adsorbed species react together (or rearrange) on the surface; neighbouring molecules are held close and correctly oriented, and bonds are already partially broken, so the activation energy is low.
4. **Desorption**: product molecules leave the surface, freeing the active sites for new reactant molecules.



Schematic of heterogeneous catalysis: gaseous reactants adsorb onto active sites of a solid metal surface, their bonds weaken, they react on the surface, and the products desorb, regenerating the sites.

WARNING **Catalyst poisoning**

Impurities in the reactant stream can poison a heterogeneous catalyst by adsorbing strongly and irreversibly onto the active sites, blocking them. For example, sulfur compounds poison the iron catalyst of the Haber process, and lead compounds destroy the platinum of catalytic converters (which is why leaded petrol had to be phased out). Poisoned catalysts must be regenerated (e.g. by heating in a controlled atmosphere) or replaced — a major industrial cost. Promoters, by contrast, are substances (e.g. K_2O and Al_2O_3 added to iron in the Haber process) that are not catalysts themselves but increase the activity or lifetime of the catalyst, for instance by improving its surface structure.

6.4 Industrial Applications

Process	Catalyst	Reaction
Haber process	Fe (with K_2O, Al_2O_3 promoters)	$N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$ — ammonia for fertilisers
Contact process	V_2O_5 (solid)	$2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g)$ — key step in H_2SO_4 manufacture
Catalytic converter	Pt / Rh on ceramic honeycomb	$2\,CO(g) + 2\,NO(g) \longrightarrow N_2(g) + 2\,CO_2(g)$; oxidation of unburnt hydrocarbons
Hydrogenation of oils	Ni (solid)	C=C bonds reduced by H_2 — margarine manufacture

In the Haber process, N_2 and H_2 chemisorb onto the iron surface; the strong N ≡ N triple bond is weakened, allowing hydrogenation step by step before NH_3 desorbs. In the catalytic converter, the catalyst also shows **selectivity**: platinum and rhodium accelerate the desired reactions (oxidation of CO, reduction of NO) far more than competing ones. Selectivity — steering reactants towards one product among several possibilities — is one of the most valuable properties a catalyst can have.

6.5 Measuring Catalytic Activity

METHOD **Determining catalytic activity by the rate enhancement factor**

1. Measure the rate constant of the uncatalysed reaction, k_{uncat}, under controlled conditions (fixed temperature, concentrations).
2. Repeat the experiment under identical conditions with a known amount of catalyst and measure k_{cat} (e.g. from the gradient of a concentration–time curve, or from the half-life for a first-order reaction, $k = \ln 2 / t_{1/2}$).

3. Compute the **rate enhancement factor** $\frac{k_{cat}}{k_{uncat}}$.

4. Interpret with the Arrhenius equation: $\ln\left(\frac{k_{cat}}{k_{uncat}}\right) = \frac{E_a(\text{uncat}) - E_a(\text{cat})}{RT}$, which gives the reduction in activation energy produced by the catalyst (assuming the pre-exponential factor A is approximately the same for both pathways).

WORKED EXAMPLE 9 **Rate enhancement and lowering of E_a**

At 298 K, a first-order reaction has $k_{uncat} = 3.0 \times 10^{-6}\ s^{-1}$. In the presence of a catalyst, $k_{cat} = 1.5 \times 10^{-2}\ s^{-1}$ at the same temperature. (a) Calculate the rate enhancement factor. (b) Calculate the reduction in activation energy. Take $R = 8.31\ J\,K^{-1}\,mol^{-1}$.

SOLUTION

(a) The enhancement factor is

$$\frac{k_{cat}}{k_{uncat}} = \frac{1.5 \times 10^{-2}}{3.0 \times 10^{-6}} = 5.0 \times 10^{3}.$$

The catalyst makes the reaction 5000 times faster.

(b) From the Arrhenius equation, assuming the pre-exponential factor A is similar for both pathways:

$$\ln\left(\frac{k_{cat}}{k_{uncat}}\right) = \frac{E_a(\text{uncat}) - E_a(\text{cat})}{RT}$$

$$\Delta E_a = RT\ln(5.0 \times 10^3) = 8.31 \times 298 \times \ln(5000)$$

$$= 8.31 \times 298 \times 8.52 \approx 2.11 \times 10^4\ J\,mol^{-1}.$$

$$\boxed{\Delta E_a \approx 21\ kJ\,mol^{-1}}$$

A reduction of only about $21\ kJ\,mol^{-1}$ in activation energy is enough to accelerate the reaction by a factor of 5000 — a striking demonstration of the exponential sensitivity of rate to E_a.

EXERCISE 6 **Catalysis in the laboratory and industry**

1. Explain, using an energy profile, why a catalyst increases the rate of the decomposition of hydrogen peroxide, $2\,H_2O_2(aq) \longrightarrow 2\,H_2O(l) + O_2(g)$, but does not change the total volume of oxygen finally obtained.

2. Classify each of the following as homogeneous or heterogeneous catalysis: (a) MnO_2 powder added to H_2O_2 solution; (b) H^+ in ester hydrolysis; (c) iron in the Haber process; (d) Fe^{2+} in the $S_2O_8^{2-}/I^-$ reaction.

3. State two reasons why the catalyst in a catalytic converter is spread as a thin coating over a ceramic honeycomb of very large surface area.

4. A catalyst lowers E_a from 75 to 50 kJ mol^{-1} at 300 K. Estimate the rate enhancement factor. (R = 8.31 J K^{-1} mol^{-1}.)

SOLUTION **Exercise 1**

1. The catalyst (e.g. MnO_2) provides an alternative pathway with lower E_a, so more molecules cross the barrier per second and O_2 is evolved faster. The catalyst changes neither ΔG nor the equilibrium position, so the final amount of oxygen — fixed by the initial amount of H_2O_2 — is unchanged; only the time to reach it is shorter.
2. (a) Heterogeneous (solid MnO_2, liquid reactant). (b) Homogeneous (all in aqueous solution). (c) Heterogeneous (solid Fe, gaseous reactants). (d) Homogeneous (all aqueous).
3. (i) Heterogeneous catalysis occurs at the surface, so a large surface area provides many more active sites and a higher rate; (ii) a thin coating uses very little of the expensive platinum and rhodium, reducing cost while the honeycomb still allows exhaust gases to flow freely.
4. $\frac{k_{cat}}{k_{uncat}} = \exp\left(\frac{\Delta E_a}{RT}\right) = \exp\left(\frac{25000}{8.31 \times 300}\right) = e^{10.03} \approx 2.3 \times 10^4$. The reaction is accelerated by a factor of about 23 000.

KEY POINTS **Key points on catalysis**

- A **catalyst** speeds up a reaction by providing an alternative pathway of lower activation energy; it is regenerated and not consumed overall.
- It does not change ΔH, ΔG or K; it accelerates forward and reverse reactions equally, so equilibrium is reached faster but its position is unchanged.
- Homogeneous catalysis (same phase) works through intermediate formation; heterogeneous catalysis (different phase) works by adsorption at active sites, surface reaction, then desorption.
- Industrial cornerstones: Fe in the Haber process, V_2O_5 in the Contact process, Pt/Rh in catalytic converters.
- Catalysts can be poisoned by impurities blocking active sites, assisted by promoters,

and valued for their selectivity.

- Catalytic activity is quantified by the enhancement factor k_{cat}/k_{uncat}, linked to the lowering of E_a through the Arrhenius equation.

7 INTEGRATION ACTIVITY

8 INTEGRATION ACTIVITY: KINETIC STUDY OF THE CATALYTIC DECOMPOSITION OF HYDROGEN PEROXIDE

8.1 Learning Objectives

By the end of this integration activity, you should be able to:

- **Design** a complete kinetic experiment: choose an appropriate monitoring method, control variables, and plan the variation of concentrations and temperature;
- **Measure** and **process** raw data (gas volumes versus time) to obtain initial rates;
- **Determine** partial orders, the overall order and the rate constant of a reaction from initial-rate data and from graphical methods (half-life);
- **Construct** an Arrhenius plot ($\ln k$ vs $1/T$) and extract the activation energy E_a;
- **Interpret** results at the molecular level using collision theory and transition-state theory, and **compare** homogeneous and heterogeneous catalysts;
- **Communicate** scientific findings in a short written report, as expected in the GCE Advanced Level practical paper.

EXPERIMENT **Context: The School Dispensary Problem**

The science club of Government Bilingual High School Nkolbisson, Yaoundé, has been asked to help the school dispensary. The dispensary stocks hydrogen peroxide solution, $H_2O_2(aq)$, used as an antiseptic, but the pharmacist complains that bottles opened during the hot season "go flat" within weeks: the peroxide decomposes and no longer releases oxygen on wounds. The decomposition reaction is:

$$2\,H_2O_2(aq) \longrightarrow 2\,H_2O(l) + O_2(g)$$

The club decides to carry out a full kinetic study of this decomposition, catalysed in two ways:

- by manganese(IV) oxide, $MnO_2(s)$ (a heterogeneous catalyst, insoluble in water);
- by iron(III) chloride solution, $FeCl_3(aq)$ providing $Fe^{3+}(aq)$ ions (a homogeneous catalyst).

The aims are to establish the rate law, measure the activation energy with each catalyst, and advise the pharmacist on storage conditions.
Laboratory equipment provided: a 100 cm^3 gas syringe connected to a conical flask, a stopwatch, a thermostatted water bath (20 °C to 60 °C), standard 1.0 mol L^{-1} H_2O_2 solu-

tion, MnO_2 powder, 0.10 mol L^{-1} $FeCl_3$ solution, distilled water and the usual glassware.

8.2 Experimental Programme

EXPERIMENT **Experiment 1 – Choice of Monitoring Method**

The club rejects titrimetric sampling (the reaction is too fast at room temperature to quench and titrate conveniently) and chooses to follow the volume of $O_2(g)$ evolved with the gas syringe.

EXPERIMENT **Experiment 2 – Initial Rates at 25 °C with MnO_2**

A fixed mass of 0.20 g of MnO_2 is used in each run; total volume is kept at 50 cm^3 by adding distilled water. Initial rates are obtained from the tangent at $t = 0$ on the volume–time curves.

Run	$[H_2O_2]$ / mol L^{-1}	mass MnO_2 / g	Initial rate / cm^3 min^{-1}
1	0.20	0.20	12.0
2	0.40	0.20	24.0
3	0.60	0.20	36.0
4	0.40	0.40	48.0

EXPERIMENT **Experiment 3 – Temperature Dependence of the Rate Constant**

Using $[H_2O_2] = 0.40$ mol L^{-1}, the initial rate is measured at three temperatures with each catalyst. The reaction is first order in H_2O_2 in both cases, so the rate constant k (units s^{-1}) is deduced from the initial rate $v_0 = k[H_2O_2]_0$.

T / K	k (MnO_2) / s^{-1}	k (Fe^{3+}) / s^{-1}
298	1.0×10^{-3}	2.0×10^{-4}
308	2.2×10^{-3}	5.0×10^{-4}
318	4.6×10^{-3}	1.2×10^{-3}

EXPERIMENT **Experiment 4 – Half-Life Check with Fe^{3+} at 298 K**

Starting from $[H_2O_2]_0$ = 0.40 mol L^{-1}, the club finds that the concentration falls to 0.20 mol L^{-1} after 58 min and to 0.10 mol L^{-1} after a further 58 min.
Data: R = 8.314 J K^{-1} mol^{-1}; molar volume of gas at 25 °C: V_m = 24.0 dm^3 mol^{-1}.

8.3 Tasks

1. Explain why the gas syringe method is appropriate here, state two precautions needed to obtain reliable volumes, and identify two variables that must be kept constant in Experiment 2.
2. Convert the initial rate of Run 1 into mol L^{-1} s^{-1} of H_2O_2 decomposed.
3. Using Runs 1–3, determine the partial order with respect to H_2O_2. Using Runs 2 and 4, determine the partial order with respect to the MnO_2 catalyst. Write the rate equation and state the overall order.
4. Calculate the rate constant k at 298 K for the Fe^{3+}-catalysed reaction from the half-life data of Experiment 4, and check its consistency with the table of Experiment 3.
5. Plot (or describe the construction of) the Arrhenius graph ln k versus $1/T$ for each catalyst, and calculate the activation energy E_a for each.
6. Using transition-state theory, explain why MnO_2 gives a larger rate constant than Fe^{3+} at the same temperature, and state what can be said about $\Delta H^{\ddagger}$ for the two catalysed pathways.
7. Classify each catalyst as homogeneous or heterogeneous, give one advantage and one disadvantage of each type in this context, and write a short recommendation to the pharmacist on how to store the H_2O_2 stock.

8.4 Model Answers with Examiner's Commentary

WORKED EXAMPLE 10 **Task 1: Choice of Method, Precautions and Controlled Variables**

Statement: Explain why the gas syringe method is appropriate here, state two precautions needed to obtain reliable volumes, and identify two variables that must be kept constant in Experiment 2.

SOLUTION

Detailed Solution:
Why the gas syringe is appropriate: The reaction produces a gaseous product, O_2, whose volume is directly proportional to the extent of reaction (amount of H_2O_2 decomposed). The gas syringe provides a continuous, non-invasive, rapid measurement without the

need for sampling and quenching, which is essential for a reaction that is relatively fast at room temperature (especially when catalysed). It directly yields $V(t)$ curves from which initial rates (tangent at $t = 0$) are easily obtained.
Two essential precautions for reliable volumes:

1. Gas-tight apparatus: Check all connections (bung, tubing, syringe barrel) for leaks before each run. A leak leads to systematically low volumes and underestimation of the rate.

2. Temperature equilibration and reading: The gas in the syringe must be at the same temperature as the water bath (or room temperature) when the volume is read. Allow the syringe to equilibrate; read the volume at eye level, at the bottom of the meniscus (or the inner edge of the plunger), and record the temperature. If the gas is warmer than the surroundings, the volume will be overestimated (Charles's law).

Additional good practice: Swirl the flask gently and continuously so that the MnO_2 powder remains suspended and its surface area stays constant; start the stopwatch exactly at the moment of mixing (adding the catalyst to the peroxide solution).
Two variables held constant in Experiment 2 (Runs 1–4):

1. Total volume of the reaction mixture (50 cm^3) – ensured by making up with distilled water.

2. Temperature (25 °C = 298 K) – maintained by the water bath.

(For Runs 1–3, the mass of MnO_2 is also constant at 0.20 g; for Runs 2 and 4, $[H_2O_2]$ is constant at 0.40 mol L^{-1}.)

WORKED EXAMPLE 11 **Task 2: Conversion of Initial Rate to Concentration Units**

Statement: Convert the initial rate of Run 1 into mol L^{-1} s^{-1} of H_2O_2 decomposed.

SOLUTION

Detailed Solution:
Run 1 gives an initial rate of O_2 evolution: $v_{O_2} = 12.0\ cm^3\ min^{-1}$.
Step 1: Convert volume rate to molar rate of O_2. Molar volume at 25 °C: $V_m = 24.0\ dm^3\ mol^{-1} = 24000\ cm^3\ mol^{-1}$.

$$\frac{dn_{O_2}}{dt} = \frac{12.0\ cm^3.min^{-1}}{24000\ cm^3.mol^{-1}} = 5.00 \times 10^{-4}\ mol.min^{-1}.$$

Step 2: Use stoichiometry to get molar rate of H_2O_2 decomposition. From the balanced equation: $2\,H_2O_2 \longrightarrow 2\,H_2O + O_2$, 2 mol H_2O_2 produce 1 mol O_2.

$$-\frac{dn_{H_2O_2}}{dt} = 2 \times \frac{dn_{O_2}}{dt} = 2 \times 5.00 \times 10^{-4} = 1.00 \times 10^{-3}\ mol.min^{-1}.$$

Step 3: Convert to per second.

$$-\frac{dn_{H_2O_2}}{dt} = \frac{1.00 \times 10^{-3}}{60} = 1.667 \times 10^{-5}\ \text{mol.s}^{-1}.$$

Step 4: Divide by solution volume to obtain concentration rate. Total volume V = 50 cm^3 = 0.0500 dm^3 = 0.0500 L.

$$v_0 = -\frac{1}{V}\frac{dn_{H_2O_2}}{dt} = \frac{1.667 \times 10^{-5}\ \text{mol.s}^{-1}}{0.0500\ \text{L}} = 3.33 \times 10^{-4}\ \text{mol.L}^{-1}.\text{s}^{-1}.$$

Answer: The initial rate of decomposition of H_2O_2 in Run 1 is **3.33 × 10^{-4}** mol L^{-1} s^{-1} (3 significant figures).

WORKED EXAMPLE 12 **Task 3: Determination of Orders and Rate Equation**

Statement: Using Runs 1–3, determine the partial order with respect to H_2O_2. Using Runs 2 and 4, determine the partial order with respect to the MnO_2 catalyst. Write the rate equation and state the overall order.

SOLUTION

Detailed Solution:
Order with respect to H_2O_2 (Runs 1, 2, 3): Mass of MnO_2 constant (0.20 g), T constant.

- Run 1 → Run 2: $[H_2O_2]$ doubles (0.20 → 0.40 mol L^{-1}); initial rate doubles (12.0 → 24.0 cm^3 min^{-1}).
- Run 1 → Run 3: $[H_2O_2]$ triples (0.20 → 0.60 mol L^{-1}); initial rate triples (12.0 → 36.0 cm^3 min^{-1}).

Rate ∝ $[H_2O_2]^1$. **Partial order = 1 (first order)** in H_2O_2.
Order with respect to MnO_2 (Runs 2 and 4): $[H_2O_2]$ constant (0.40 mol L^{-1}), T constant.

- Run 2 → Run 4: mass of MnO_2 doubles (0.20 → 0.40 g); initial rate doubles (24.0 → 48.0 cm^3 min^{-1}).

Rate ∝ (mass of catalyst)^1. Since the catalyst is a solid, its "concentration" is proportional to its mass (or surface area) for a given particle size. **Partial order = 1 (first order)** in catalyst.
Rate equation:

$v = k\,[H_2O_2]^1\,[\text{catalyst}]^1$ or $v = k'\,[H_2O_2]$ with $k' = k[\text{catalyst}]$ (pseudo-first-order constant).

Overall order = 1 + 1 = **2** (second order overall).
Consistency check: With a fixed amount of catalyst, the reaction behaves as pseudo-first order in H_2O_2. This is confirmed by Experiment 4 (constant half-life).

WORKED EXAMPLE 13 **Task 4: Rate Constant from Half-Life Data**

Statement: Calculate the rate constant k at 298 K for the Fe^{3+}-catalysed reaction from the half-life data of Experiment 4, and check its consistency with the table of Experiment 3.

SOLUTION

Detailed Solution:
Experiment 4 shows: $[H_2O_2]$ falls from 0.40 mol L^{-1} to 0.20 mol L^{-1} in 58 min, and from 0.20 mol L^{-1} to 0.10 mol L^{-1} in a further 58 min. The half-life $t_{1/2}$ is constant (58 min) independent of concentration. This is the hallmark of a **first-order** reaction in H_2O_2.
For a first-order reaction:

$$k = \frac{\ln 2}{t_{1/2}}.$$

Convert $t_{1/2}$ to seconds: $t_{1/2}$ = 58 min × 60 s.min^{-1} = 3480 s.

$$k = \frac{0.6931}{3480\ \text{s}} = 1.99 \times 10^{-4}\ \text{s}^{-1} \approx \mathbf{2.0 \times 10^{-4}\ s^{-1}}.$$

Consistency check: The table in Experiment 3 gives $k(Fe^{3+})$ at 298 K = **2.0×10^{-4} s^{-1}**. The two values are in exact agreement, confirming the first-order kinetics and the reliability of the measurements.

WORKED EXAMPLE 14 **Task 5: Arrhenius Plots and Activation Energies**

Statement: Plot (or describe the construction of) the Arrhenius graph $\ln k$ versus $1/T$ for each catalyst, and calculate the activation energy E_a for each.

SOLUTION

Detailed Solution:
The Arrhenius equation in linear form:

$$\ln k = \ln A - \frac{E_a}{R}\frac{1}{T}.$$

A plot of $\ln k$ (y-axis) against $1/T$ (x-axis) yields a straight line of slope $m = -E_a/R$.
Step 1: Tabulate $\ln k$ and $1/T$.

T / K	$10^3/T$ / K^{-1}	$\ln k$ (MnO_2)	$\ln k$ (Fe^{3+})
298	3.356	$\ln(1.0 \times 10^{-3}) = -6.908$	$\ln(2.0 \times 10^{-4}) = -8.517$
308	3.247	$\ln(2.2 \times 10^{-3}) = -6.119$	$\ln(5.0 \times 10^{-4}) = -7.601$
318	3.145	$\ln(4.6 \times 10^{-3}) = -5.381$	$\ln(1.2 \times 10^{-3}) = -6.725$

Step 2: Determine slopes (using the two extreme points for best accuracy).

For MnO_2:

$$m_{MnO_2} = \frac{\ln k_{318} - \ln k_{298}}{1/318 - 1/298} = \frac{-5.381 - (-6.908)}{3.145 \times 10^{-3} - 3.356 \times 10^{-3}} = \frac{1.527}{-0.211 \times 10^{-3}} = -7237\ \text{K}.$$

$$E_a(MnO_2) = -m \times R = 7237 \times 8.314 = 60160\ \text{J mol}^{-1} \approx \mathbf{60.2\ kJ.mol^{-1}}.$$

For Fe^{3+}:

$$m_{Fe^{3+}} = \frac{-6.725 - (-8.517)}{3.145 \times 10^{-3} - 3.356 \times 10^{-3}} = \frac{1.792}{-0.211 \times 10^{-3}} = -8493\ \text{K}.$$

$$E_a(Fe^{3+}) = 8493 \times 8.314 = 70600\ \text{J mol}^{-1} \approx \mathbf{70.6\ kJ.mol^{-1}}.$$

Note: In an examination, you would plot the points on graph paper, draw the best-fit straight line, and measure the slope using a large triangle. The values above are the precise calculations from the given data; graphical determination typically yields $E_a(MnO_2) \approx 58\text{--}60\ \text{kJ mol}^{-1}$ and $E_a(Fe^{3+}) \approx 68\text{--}71\ \text{kJ mol}^{-1}$.

KEY POINTS **Arrhenius Plot – Examination Technique**

- Always convert T to Kelvin and k to consistent units (s^{-1} here).
- Calculate $1/T$ in K^{-1} (often $\times 10^3$ for convenient scaling).
- Plot $\ln k$ vs $1/T$; the slope is negative.
- E_a = -slope × R. Give E_a in kJ mol^{-1} (divide by 1000).
- The intercept gives $\ln A$ (pre-exponential factor), but this is rarely asked at A Level.

WORKED EXAMPLE 15 **Task 6: Transition-State Theory Interpretation**

Statement: Using transition-state theory, explain why MnO_2 gives a larger rate constant than Fe^{3+} at the same temperature, and state what can be said about $\Delta H^{\ddagger}$ for the two catalysed pathways.

SOLUTION

Detailed Solution:
In Transition-State Theory (TST), the rate constant is given by the Eyring equation:

$$k = \frac{k_B T}{h} e^{-\Delta G^{\ddagger}/RT} = \frac{k_B T}{h} e^{\Delta S^{\ddagger}/R} e^{-\Delta H^{\ddagger}/RT},$$

where $\Delta G^{\ddagger} = \Delta H^{\ddagger} - T\Delta S^{\ddagger}$ is the Gibbs free energy of activation.

Comparison at 298 K: $k(MnO_2) = 1.0 \times 10^{-3}\ s^{-1}$; $k(Fe^{3+}) = 2.0 \times 10^{-4}\ s^{-1}$. $k(MnO_2)$ is 5 times larger than $k(Fe^{3+})$. Therefore, the MnO_2-catalysed pathway has a lower $\Delta G^{\ddagger}$ (more negative $\Delta G^{\ddagger}$ means faster reaction).
Enthalpy of activation $\Delta H^{\ddagger}$: For reactions in solution, $E_a \approx \Delta H^{\ddagger} + RT$ (the RT term is small, $\approx 2.5\ kJ\ mol^{-1}$ at 298 K). Thus E_a is a good approximation for $\Delta H^{\ddagger}$. From Task 5:

- $E_a(MnO_2) \approx 60\ kJ\ mol^{-1} \Rightarrow \Delta H^{\ddagger}_{MnO_2} \approx 58\ kJ\ mol^{-1}$.
- $E_a(Fe^{3+}) \approx 71\ kJ\ mol^{-1} \Rightarrow \Delta H^{\ddagger}_{Fe^{3+}} \approx 68\ kJ\ mol^{-1}$.

$\Delta H^{\ddagger}$ is significantly lower (by $\approx 10\ kJ\ mol^{-1}$) for the MnO_2 pathway.
Conclusion: The MnO_2 surface stabilises the activated complex (transition state) more effectively than the Fe^{3+} ions in solution, lowering the enthalpy barrier. The entropy of activation $\Delta S^{\ddagger}$ may also play a role: on a surface, reactants can be pre-adsorbed in a favourable orientation (less negative $\Delta S^{\ddagger}$), further reducing $\Delta G^{\ddagger}$. However, the dominant factor here is the lower $\Delta H^{\ddagger}$.

WORKED EXAMPLE 16 **Task 7: Catalyst Classification, Pros/Cons and Recommendation**

Statement: Classify each catalyst as homogeneous or heterogeneous, give one advantage and one disadvantage of each type in this context, and write a short recommendation to the pharmacist on how to store the H_2O_2 stock.

SOLUTION

Detailed Solution:

Catalyst	Classification	Advantage / Disadvantage in this context
$MnO_2(s)$	Heterogeneous (solid catalyst, liquid reactants)	Adv: Easily separated by filtration, can be reused indefinitely (cost-effective). Disadv: Activity depends on surface area (particle size); can be poisoned by impurities adsorbing on active sites; mass transfer limitations possible.
$Fe^{3+}(aq)$	Homogeneous (same phase – aqueous)	Adv: Uniform mixing, every catalytic ion is accessible; no mass transfer limitations; often higher selectivity. Disadv: Cannot be separated from the product (contaminates the H_2O_2 solution); difficult to recover/reuse; here it is less active (higher E_a, lower k).

Recommendation to the Pharmacist:

> "Store the hydrogen peroxide stock in a cool, dark cupboard (ideally below 20 °C), in tightly closed, opaque (brown glass) bottles. Low temperature exponentially decreases the rate constant (Arrhenius law); light (especially UV) accelerates photodecomposition. Keep bottles away from dust, rust, and metal ions (Fe, Cu, Mn) which act as homogeneous or heterogeneous catalysts. During the hot season in Yaoundé, consider refrigeration if possible, and avoid leaving bottles open. Replace stock regularly as decomposition is inevitable over time."

DID YOU KNOW? Real-World Context

Commercial H_2O_2 solutions contain stabilisers (e.g., phosphoric acid, stannate) that chelate metal ions and slow decomposition. The "fizzing" on a wound is due to catalase enzyme in blood/tissues – a biological catalyst far more efficient than MnO_2 or Fe^{3+}!

KEY POINTS Key Takeaways for the GCE Advanced Level

- Gas syringe method: Ideal for gas-evolving reactions; continuous data; initial rate =

tangent at $t = 0$.

- Initial rates method: Compare rates where only one concentration changes; order n from $(rate_2/rate_1) = ([A]_2/[A]_1)^n$.
- Half-life method: Constant $t_{1/2}$ ⇒ first order; $t_{1/2} \propto 1/[A]_0$ ⇒ second order.
- Arrhenius plot: $\ln k$ vs $1/T$ straight line; slope = $-E_a/R$; E_a in kJ mol^{-1}.
- Transition-State Theory: $k \propto e^{-\Delta G^{\ddagger}/RT}$; lower $\Delta H^{\ddagger}$ (or more positive $\Delta S^{\ddagger}$) ⇒ larger k.
- Catalysis: Homogeneous = same phase (hard to separate); Heterogeneous = different phase (easy to separate, surface area matters).

WARNING **Common Pitfalls in Kinetics Questions**

- Forgetting to convert time to seconds when calculating k (units of k depend on overall order).
- Using V_m at STP (22.4 dm^3) instead of the given molar volume at the experimental temperature.
- Confusing order with stoichiometric coefficient (they are unrelated for complex reactions).
- Plotting k vs T instead of $\ln k$ vs $1/T$ for Arrhenius.
- Stating that a catalyst "lowers the activation energy" without mentioning it provides an alternative reaction pathway with a lower E_a.
- Not specifying that for a heterogeneous catalyst, the rate depends on surface area (mass for fixed particle size), not "concentration".

9 METHODS AND WORKED EXAMPLES

9.1 Methods and Worked Examples

METHOD **Determining Reaction Rates from Concentration–Time Data**

1. Identify the data type: You may be given a table of concentration versus time, a graph, or a description of an experiment (gas volume, absorbance, conductivity, pressure).
2. Plot the graph: If tabular data are given, plot concentration (or a quantity proportional to it: volume, absorbance, pressure) on the y-axis against time on the x-axis. Draw a smooth curve through the points.
3. Average rate over Δt: $\text{Rate}_{\text{avg}} = -\dfrac{\Delta[\text{Reactant}]}{\Delta t} = -\dfrac{[\text{R}]_{t_2} - [\text{R}]_{t_1}}{t_2 - t_1}$ (the minus sign makes the rate positive, since reactant concentration decreases). For products, $\text{Rate}_{\text{avg}} = +\dfrac{\Delta[\text{Product}]}{\Delta t}$.
4. Instantaneous rate at time t: Draw a tangent to the curve at time t with a ruler. Calculate its gradient: $\text{Rate}_{\text{inst}} = \dfrac{\text{change in concentration}}{\text{change in time}}$ = gradient of tangent.
5. Initial rate ($t = 0$): Draw the tangent at $t = 0$. This is the most reliable rate for comparing experiments with different initial concentrations, because at $t = 0$ the concentrations are known exactly and no products have accumulated.
6. Units: Always express rate in $\text{mol dm}^{-3}\,\text{s}^{-1}$. Convert time to seconds if given in minutes, and volumes to dm^3 if given in cm^3.
7. Stoichiometry link: For $aA + bB \longrightarrow cC + dD$,

$$\text{Rate} = -\frac{1}{a}\frac{d[\text{A}]}{dt} = -\frac{1}{b}\frac{d[\text{B}]}{dt} = +\frac{1}{c}\frac{d[\text{C}]}{dt} = +\frac{1}{d}\frac{d[\text{D}]}{dt}.$$

WORKED EXAMPLE 17 **Rate Determination from Gas Evolution Data**

In an experiment, $\text{CaCO}_3(\text{s})$ reacts with excess HCl(aq): $\text{CaCO}_3(\text{s}) + 2\,\text{HCl(aq)} \longrightarrow \text{CaCl}_2(\text{aq}) + \text{H}_2\text{O(l)} + \text{CO}_2(\text{g})$. The volume of CO_2 collected at room temperature and pressure (RTP, molar volume = $24.0\ \text{dm}^3\ \text{mol}^{-1}$) is recorded:

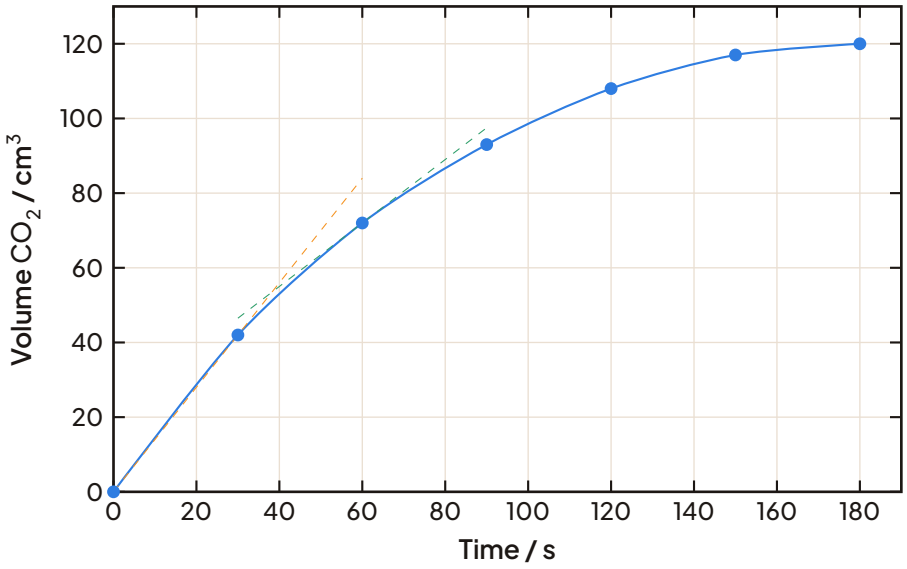
Time (s)	0	30	60	90	120	150	180
Vol CO_2 (cm^3)	0	42	72	93	108	117	120

(a) Plot the graph of volume against time.

(b) Determine the initial rate of reaction in mol dm^{-3} s^{-1} if the volume of the reaction mixture is 50.0 cm^3.
(c) Determine the instantaneous rate at t = 60 s.
(d) Explain why the rate decreases with time.

SOLUTION

Detailed Solution
(a) Graph: Plot Volume (cm^3) on the y-axis, Time (s) on the x-axis. Points: (0, 0), (30, 42), (60, 72), (90, 93), (120, 108), (150, 117), (180, 120). Draw a smooth curve starting steep and levelling off at 120 cm^3 (all the $CaCO_3$ has reacted).



Volume of CO_2 against time. Orange dashed line: tangent at t = 0 (initial rate). Green dashed line: tangent at t = 60 s (instantaneous rate).

(b) Initial Rate: Gradient of the tangent at t = 0. Using the first two points as an approximation of the tangent:

$$\text{Gradient} \approx \frac{42 - 0}{30 - 0} = 1.40\ \text{cm}^3\ \text{s}^{-1}.$$

Convert to $dm^3\ s^{-1}$: $1.40\ cm^3\ s^{-1} = 1.40 \times 10^{-3}\ dm^3\ s^{-1}$. Moles of CO_2 per second = $\frac{1.40 \times 10^{-3}}{24.0} = 5.83 \times 10^{-5}$ mol s^{-1}. Rate of reaction (with respect to CO_2 formation, per unit volume of mixture):

$$\text{Rate} = \frac{1}{V_{\text{mixture}}} \times \frac{dn_{CO_2}}{dt} = \frac{5.83 \times 10^{-5}}{50.0 \times 10^{-3}} = \boxed{1.17 \times 10^{-3}\ \text{mol}\ \text{dm}^{-3}\ \text{s}^{-1}}.$$

(The rate of disappearance of HCl would be $-\frac{1}{2}\frac{d[HCl]}{dt}$ equal to this same value, since 2 mol of HCl are consumed per 1 mol of CO_2 formed.)
(c) Instantaneous Rate at t = 60 s: Draw the tangent at t = 60 s. Estimate its gradient using two points on the tangent, e.g. at t = 30 s ($V \approx 46.5\ cm^3$ on the tangent) and t = 90 s

($V \approx 97.5\ \text{cm}^3$):

$$\text{Gradient} \approx \frac{97.5 - 46.5}{90 - 30} = \frac{51}{60} = 0.85\ \text{cm}^3\ \text{s}^{-1}.$$

$$\text{Moles per second} = \frac{0.85 \times 10^{-3}}{24.0} = 3.54 \times 10^{-5}\ \text{mol s}^{-1}.$$

$$\text{Rate} = \frac{3.54 \times 10^{-5}}{0.0500} = \boxed{7.1 \times 10^{-4}\ \text{mol dm}^{-3}\ \text{s}^{-1}}.$$

(d) Reason for the decrease: The rate depends on the concentration of HCl (the reactant in solution). As the reaction proceeds, HCl is consumed and its concentration falls. Since Rate = $k[\text{HCl}]^n$, a lower concentration gives a lower rate, so the curve becomes less steep. The rate falls to zero when the $CaCO_3$ is used up (the acid is in excess).

METHOD **Titrimetric (Chemical) Method for Rate Determination**

1. Principle: Samples (aliquots) of the reaction mixture are withdrawn at measured times and the reaction in each sample is quenched (stopped or drastically slowed) by, for example, adding ice-cold water, adding a reagent that removes a catalyst, or neutralising an acid.
2. Titration: The quenched sample is titrated against a standard solution to find the concentration of a reactant remaining or product formed (e.g. H^+, I_2, H_2O_2).
3. Data processing: The titre V_t is proportional to the concentration of the titrated species at time t. Plot V_t against t, or test integrated rate laws (e.g. $\ln V_t$ against t for first order when $V_\infty = 0$).
4. Key reactions:
   - Acid-catalysed ester hydrolysis: $CH_3COOCH_3 + H_2O \overset{H^+}{\rightleftharpoons} CH_3COOH + CH_3OH$. Quench by adding the aliquot to ice-cold water; titrate the total acid with standard NaOH. (The increase in titre above the value at $t = 0$ measures the CH_3COOH formed.)
   - Propanone + iodine: $CH_3COCH_3 + I_2 \xrightarrow{H^+} CH_3COCH_2I + H^+ + I^-$. Quench with $NaHCO_3$ (neutralises the H^+ catalyst); titrate the remaining I_2 with standard $Na_2S_2O_3$ using starch indicator.
   - Decomposition of hydrogen peroxide: $2\,H_2O_2 \longrightarrow 2\,H_2O + O_2$. Quench by adding the aliquot to cold dilute acid; titrate remaining H_2O_2 with standard $KMnO_4$.
5. Advantage / disadvantage: It gives a direct chemical measure of concentration; but it is discontinuous (sampling errors), relatively slow, and quenching may not be perfectly instantaneous.

WORKED EXAMPLE 18 **Kinetics of the Acid-Catalysed Propanone–Iodine Reaction**

The reaction $CH_3COCH_3(aq) + I_2(aq) \xrightarrow{H^+} CH_3COCH_2I(aq) + H^+(aq) + I^-(aq)$ is studied with propanone and acid in large excess. 10.0 cm^3 aliquots are withdrawn at intervals, quenched with $NaHCO_3$, and titrated with 0.0100 mol dm^{-3} $Na_2S_2O_3$ ($I_2 + 2\,S_2O_3^{2-} \longrightarrow 2\,I^- + S_4O_6^{2-}$).

Time (min)	0	5	10	15	20	∞
Titre (cm^3)	25.0	18.4	13.5	9.9	7.3	0.0

(a) Show that the reaction is first order with respect to I_2.
(b) Calculate the rate constant k (in s^{-1}).
(c) Determine the half-life $t_{1/2}$.
(d) Why is the order with respect to CH_3COCH_3 and H^+ not determined by this experiment?

SOLUTION

Detailed Solution
(a) First-order proof: For a first-order reaction, $\ln[A]_t = \ln[A]_0 - kt$. Here the titre $V_t \propto [I_2]_t$, and at $t = \infty$, $V_\infty = 0$ (all the I_2 has reacted). Hence $[I_2]_t \propto V_t$ and the integrated rate law becomes $\ln V_t = \ln V_0 - kt$. Calculate $\ln(V_t)$:

t (min)	0	5	10	15	20
V_t (cm^3)	25.0	18.4	13.5	9.9	7.3
$\ln(V_t)$	3.219	2.912	2.603	2.293	1.988

A plot of $\ln(V_t)$ against t is a straight line of negative slope, confirming first order. Alternatively, check the constancy of $k = \frac{1}{t}\ln\frac{V_0}{V_t}$:

$$t = 5: \quad k = \tfrac{1}{5}\ln\tfrac{25.0}{18.4} = 0.0614\ \text{min}^{-1}$$
$$t = 10: \quad k = \tfrac{1}{10}\ln\tfrac{25.0}{13.5} = 0.0616\ \text{min}^{-1}$$
$$t = 15: \quad k = \tfrac{1}{15}\ln\tfrac{25.0}{9.9} = 0.0618\ \text{min}^{-1}$$
$$t = 20: \quad k = \tfrac{1}{20}\ln\tfrac{25.0}{7.3} = 0.0616\ \text{min}^{-1}$$

k is constant ($\approx 0.0616\ \text{min}^{-1}$). The reaction is first order with respect to I_2.
(b) Rate constant in s^{-1}:

$$k = 0.0616\ \text{min}^{-1} = \frac{0.0616}{60}\ \text{s}^{-1} = \boxed{1.03 \times 10^{-3}\ \text{s}^{-1}}.$$

(c) Half-life: For first order, $t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{0.0616\ \text{min}^{-1}} = \boxed{11.3\ \text{min}}$ (≈ 675 s).
(d) Reason: CH_3COCH_3 and H^+ are in large excess compared with I_2, so their concentrations remain effectively constant during the reaction. The observed rate law is Rate =

$k'[I_2]$ where $k' = k[CH_3COCH_3]^x[H^+]^y$ (a pseudo-first-order constant). To find x and y, the experiment must be repeated varying the initial $[CH_3COCH_3]$ and $[H^+]$ separately while keeping the other reagents in excess.

METHOD **Determining Order of Reaction: Initial Rates Method**

1. Write the rate law: Rate = $k[A]^x[B]^y$. Overall order = $x + y$.
2. Select experiments: Compare two experiments in which only one reactant concentration changes (all others constant).
3. Calculate the ratio: $\frac{\text{Rate}_2}{\text{Rate}_1} = \left(\frac{[A]_2}{[A]_1}\right)^x$.
4. Solve for the order x: Take logarithms: $x = \frac{\log(\text{Rate}_2/\text{Rate}_1)}{\log([A]_2/[A]_1)}$. Usually the ratios are simple (doubling the concentration multiplies the rate by 1, 2, 4 or 8), giving integer orders 0, 1, 2 or 3.
5. Repeat for the other reactants.
6. Calculate k: Substitute the orders and the data from any one experiment into the rate law. State the units of k according to the overall order:
   - Order 0: $\text{mol dm}^{-3}\text{ s}^{-1}$
   - Order 1: s^{-1}
   - Order 2: $\text{mol}^{-1}\text{ dm}^3\text{ s}^{-1}$
   - Order 3: $\text{mol}^{-2}\text{ dm}^6\text{ s}^{-1}$
7. Check: Use k to predict the rate for another experiment and compare with the measured value.

WORKED EXAMPLE 19 **Initial Rates Analysis: Reaction between NO and O2**

The reaction $2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2(g)$ is studied at constant temperature. Initial rates are measured:

Exp.	$[NO]_0$ (mol dm^{-3})	$[O_2]_0$ (mol dm^{-3})	Initial Rate (mol dm^{-3} s^{-1})
1	0.010	0.010	2.5×10^{-5}
2	0.020	0.010	1.0×10^{-4}
3	0.010	0.020	5.0×10^{-5}
4	0.020	0.020	2.0×10^{-4}

(a) Determine the order with respect to NO and O_2. Write the rate equation.
(b) Calculate the rate constant k with units.
(c) Predict the initial rate for Experiment 5: $[NO]_0 = 0.030$ mol dm^{-3}, $[O_2]_0 = 0.015$ mol dm^{-3}.

SOLUTION

Detailed Solution
(a) Order with respect to NO (compare Experiments 1 and 2): $[O_2]$ is constant (0.010); [NO] doubles (0.010 → 0.020). Rate ratio: $\frac{1.0 \times 10^{-4}}{2.5 \times 10^{-5}} = 4$. $\left(\frac{0.020}{0.010}\right)^x = 2^x = 4 \implies x = 2$. Second order with respect to NO.
Order with respect to O_2 (compare Experiments 1 and 3): [NO] is constant (0.010); $[O_2]$ doubles (0.010 → 0.020). Rate ratio: $\frac{5.0 \times 10^{-5}}{2.5 \times 10^{-5}} = 2$. $\left(\frac{0.020}{0.010}\right)^y = 2^y = 2 \implies y = 1$. First order with respect to O_2.
Rate equation: Rate = $k[NO]^2[O_2]$. Overall order = 2 + 1 = 3.
(b) Rate constant k: Use Experiment 1:

$$2.5 \times 10^{-5} = k(0.010)^2(0.010) = k \times 1.0 \times 10^{-6} \implies k = \frac{2.5 \times 10^{-5}}{1.0 \times 10^{-6}} = 25.$$

Units: $\frac{\text{mol dm}^{-3}\,\text{s}^{-1}}{(\text{mol dm}^{-3})^3} = \text{mol}^{-2}\,\text{dm}^6\,\text{s}^{-1}$.

$$\boxed{k = 25\ \text{mol}^{-2}\,\text{dm}^6\,\text{s}^{-1}}.$$

Check with Experiment 4: predicted rate = $25 \times (0.020)^2 \times (0.020) = 25 \times 8.0 \times 10^{-6} = 2.0 \times 10^{-4}$ mol dm^{-3} s^{-1}. This matches the measured value, confirming the rate law.
(c) Prediction for Experiment 5:

$$\text{Rate} = 25 \times (0.030)^2 \times (0.015) = 25 \times 9.0 \times 10^{-4} \times 1.5 \times 10^{-2} = \boxed{3.4 \times 10^{-4}\ \text{mol dm}^{-3}\,\text{s}^{-1}}.$$

METHOD **Graphical Determination of Order: Integrated Rate Laws and Half-Life**

1. Identify the measured variable: concentration [A], titre V_t, pressure P_t, absorbance A_t, conductivity κ_t. Ensure it is proportional to the concentration of one species.

2. Test for zero order: Plot $[A]_t$ against t. A straight line $\Longrightarrow$ zero order. Slope = $-k$. $t_{1/2} = \frac{[A]_0}{2k}$ (depends on $[A]_0$).
3. Test for first order: Plot $\ln[A]_t$ (or $\ln V_t$, or $\ln(V_t - V_\infty)$) against t. A straight line $\Longrightarrow$ first order. Slope = $-k$. $t_{1/2} = \frac{\ln 2}{k}$ (constant, independent of $[A]_0$).
4. Test for second order: Plot $\frac{1}{[A]_t}$ against t. A straight line $\Longrightarrow$ second order. Slope = $+k$. $t_{1/2} = \frac{1}{k[A]_0}$ (halves when $[A]_0$ doubles).
5. Half-life method (experimental): Measure $t_{1/2}$ for different initial concentrations $[A]_0$:
   - $t_{1/2}$ constant ⇒ first order;
   - $t_{1/2} \propto [A]_0$ ⇒ zero order;
   - $t_{1/2} \propto 1/[A]_0$ ⇒ second order.
6. Choose the plot giving the best straight line.

WORKED EXAMPLE 20 **Graphical Method: Decomposition of N_2O_5**

The gas-phase decomposition $2\,N_2O_5(g) \longrightarrow 4\,NO_2(g) + O_2(g)$ is followed by measuring the total pressure P_{tot} in a constant-volume vessel at 338 K. Initial pressure P_0 = 100 kPa (pure N_2O_5).

Time (s)	0	200	400	600	800	1000
P_{tot} (kPa)	100	118	132	143	152	159

(a) Derive the relationship between $P_{N_2O_5}$ and P_{tot}.
(b) Show that the reaction is first order with respect to N_2O_5 and determine k.
(c) Calculate the half-life.
(d) What is the partial pressure of O_2 at t = 600 s?

SOLUTION

Detailed Solution
(a) Stoichiometry and pressure: Let x = partial pressure of N_2O_5 that has decomposed at time t. From the equation $2\,N_2O_5 \longrightarrow 4\,NO_2 + O_2$, 2 mol of N_2O_5 give 4 mol of NO_2 and 1 mol of O_2. Hence, in pressure units:

	N_2O_5	NO_2	O_2
Initial	100	0	0
Change	$-x$	$+2x$	$+\frac{x}{2}$
At time t	$100 - x$	$2x$	$\frac{x}{2}$

$$P_{tot} = (100 - x) + 2x + \frac{x}{2} = 100 + \frac{3x}{2} \implies x = \frac{2}{3}(P_{tot} - 100).$$

Therefore $\boxed{P_{N_2O_5} = 100 - \frac{2}{3}(P_{tot} - 100)}$.

(b) First-order test: Calculate $P_{N_2O_5}$ and $\ln(P_{N_2O_5})$:

t (s)	0	200	400	600	800	1000
P_{tot} (kPa)	100	118	132	143	152	159
$P_{N_2O_5}$ (kPa)	100	88.0	78.7	71.3	65.3	60.7
$\ln(P_{N_2O_5})$	4.605	4.477	4.365	4.267	4.179	4.106

(Example: at t = 200 s, $P_{N_2O_5} = 100 - \frac{2}{3}(118 - 100) = 100 - 12.0 = 88.0$ kPa.)
A plot of $\ln(P_{N_2O_5})$ against t is a straight line, confirming first order. Slope (using t = 0 and t = 1000 s):

$$\text{Slope} = \frac{4.106 - 4.605}{1000 - 0} = -4.99 \times 10^{-4}\ \text{s}^{-1} \implies k = -\text{slope} = \boxed{5.0 \times 10^{-4}\ \text{s}^{-1}}.$$

(c) Half-life:

$$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{5.0 \times 10^{-4}} = \boxed{1.39 \times 10^{3}\ \text{s} (\approx 23\ \text{min})}.$$

(d) Partial pressure of O_2 at t = 600 s:

$$P_{O_2} = \frac{x}{2} = \frac{1}{2} \times \frac{2}{3}(P_{tot} - 100) = \frac{P_{tot} - 100}{3} = \frac{143 - 100}{3} = \boxed{14.3\ \text{kPa}}.$$

METHOD **Arrhenius Equation: Determining E_a and A**

1. Arrhenius equation: $k = Ae^{-E_a/RT}$, where A is the pre-exponential (frequency) factor, E_a the activation energy, R = 8.314 J mol^{-1} K^{-1}, T the absolute temperature.
2. Linear form: $\ln k = \ln A - \frac{E_a}{R} \cdot \frac{1}{T}$.
   - Plot $\ln k$ (y-axis) against $\frac{1}{T}$ (x-axis, in K^{-1}).
   - A straight line confirms Arrhenius behaviour.

- Slope $= -\dfrac{E_a}{R} \implies E_a = -\text{slope} \times R$.
- Intercept $= \ln A \implies A = e^{\text{intercept}}$.

3. Two-point form (only two temperatures available): $\ln\dfrac{k_2}{k_1} = \dfrac{E_a}{R}\left(\dfrac{1}{T_1} - \dfrac{1}{T_2}\right)$.
4. Units: E_a in J mol^{-1} (convert to kJ mol^{-1} at the end); T in kelvin; A has the same units as k.

WARNING **Common Errors with the Arrhenius Equation**

Using T in °C instead of kelvin; forgetting to convert E_a from J to kJ; confusing $\log_{10}$ with ln (if $\log_{10} k$ is plotted, the slope is $-E_a/2.303R$); quoting A without units.

WORKED EXAMPLE 21 **Activation Energy from Temperature Dependence**

The rate constant k for the decomposition of N_2O_5 was measured at different temperatures:

T (°C)	25	35	45	55	65
k (s^{-1})	1.7×10^{-5}	6.5×10^{-5}	2.4×10^{-4}	8.2×10^{-4}	2.7×10^{-3}

(a) Plot the Arrhenius graph and determine E_a (in kJ mol^{-1}) and A.
(b) Calculate k at 50 °C.
(c) Explain the physical significance of A and state why E_a is taken as independent of temperature in this model.

SOLUTION

Detailed Solution
(a) Data processing: Convert T to kelvin ($T/\text{K} = T/^\circ\text{C} + 273$). Calculate $1/T$ and ln k:

T (K)	298	308	318	328	338
$10^3/T$ (K^{-1})	3.356	3.247	3.145	3.049	2.959
k (s^{-1})	1.7×10^{-5}	6.5×10^{-5}	2.4×10^{-4}	8.2×10^{-4}	2.7×10^{-3}
ln k	−10.98	−9.64	−8.33	−7.11	−5.91

Plot ln k against $1/T$ and draw the best straight line. Select two points on the line, far apart: Point 1: $(3.356 \times 10^{-3}, -10.98)$; Point 2: $(2.959 \times 10^{-3}, -5.91)$.

$$\text{Slope} = \frac{-5.91 - (-10.98)}{(2.959 - 3.356) \times 10^{-3}} = \frac{5.07}{-3.97 \times 10^{-4}} = -1.28 \times 10^4\ \text{K}.$$

$$E_a = -\text{slope} \times R = 1.28 \times 10^4 \times 8.314 = 1.06 \times 10^5 \text{ J mol}^{-1} = \boxed{106 \text{ kJ mol}^{-1}}.$$

Intercept: using $c = y - mx$ at Point 1:

$$\ln A = -10.98 - (-1.28 \times 10^4 \times 3.356 \times 10^{-3}) = -10.98 + 42.96 = 31.98 \implies A = e^{31.98} \approx \boxed{7 \times 10^{13} \text{ s}^{-1}}.$$

(A has the same units as k, i.e. s^{-1} for a first-order reaction.)
(b) k at 50°C (323 K): $1/T = 3.096 \times 10^{-3} \text{ K}^{-1}$.

$$\ln k = 31.98 - (1.28 \times 10^4)(3.096 \times 10^{-3}) = 31.98 - 39.63 = -7.65 \implies k = e^{-7.65} = \boxed{4.8 \times 10^{-4} \text{ s}^{-1}}.$$

(Check: this lies between 2.4×10^{-4} at 45°C and 8.2×10^{-4} at 55°C — reasonable.)
(c) Significance: A (the pre-exponential factor) is related to the collision frequency and the orientation (steric) requirement: $A = pZ$, where Z is the collision frequency and p the steric factor. It represents the value k would approach at infinitely high temperature, when effectively every collision has enough energy to react. E_a is the minimum energy a colliding pair must possess for reaction to occur — the height of the energy barrier between reactants and the transition state. In the simple Arrhenius model this barrier height is a property of the potential energy surface and is therefore treated as constant (independent of T); the temperature dependence of k comes almost entirely from the exponential term $e^{-E_a/RT}$.

METHOD **Collision Theory and Transition State Theory: Key Ideas**

1. Collision theory: Rate = $Z \times f \times p$, where
   - Z: collision frequency (increases with concentration and with $\sqrt{T}$);
   - $f = e^{-E_a/RT}$: fraction of collisions with energy ≥ E_a (from the Maxwell–Boltzmann distribution);
   - p: steric factor (probability that the collision has the correct orientation).
2. Maxwell–Boltzmann distribution: the curve of molecular energies shows that only a small fraction of molecules have $E \geq E_a$; raising the temperature flattens and shifts the curve, greatly increasing this fraction — hence the strong temperature dependence of rates.
3. Transition state theory (TST): Reactants pass through a high-energy activated complex (transition state) at the top of the energy barrier: $A + B \rightleftharpoons AB^{\ddagger} \rightarrow$ products. The rate depends on the concentration of the activated complex and the frequency with which it decomposes into products.
4. Similarities: both theories require a minimum energy (E_a) for reaction; both predict the Arrhenius temperature dependence; both involve an activated, high-energy configuration of atoms.

5. Differences:

- Collision theory treats molecules as hard spheres and applies best to simple gas-phase reactions; the steric factor p is introduced empirically.
- TST considers the structure and energy of the activated complex explicitly; it applies to reactions in solution as well as gases, and accounts for orientation and entropy effects through the nature of the transition state rather than an empirical factor.

WORKED EXAMPLE 22 **Collision Theory: Fraction of Effective Collisions**

For a bimolecular gas reaction at 300 K, $E_a = 50.0\ \text{kJ mol}^{-1}$. The measured pre-exponential factor is $A = 2.0 \times 10^{11}\ \text{mol}^{-1}\ \text{dm}^3\ \text{s}^{-1}$, and the collision frequency factor Z (for unit concentrations) is calculated as $4.0 \times 10^{11}\ \text{mol}^{-1}\ \text{dm}^3\ \text{s}^{-1}$.
(a) Calculate the fraction of collisions with energy $\geq E_a$ at 300 K.
(b) Calculate the steric factor p and comment on its value.
(c) Explain, in terms of the Maxwell–Boltzmann distribution, why the rate roughly doubles when the temperature rises by about 10 K near room temperature.
(d) Why might the steric factor p be less than 1?

SOLUTION

Detailed Solution
(a) Fraction f (Maxwell–Boltzmann):

$$f = e^{-E_a/RT} = \exp\left(-\frac{50.0 \times 10^3}{8.314 \times 300}\right) = \exp(-20.05) = \boxed{2.0 \times 10^{-9}}.$$

Only about 2 collisions in every 10^9 have sufficient energy to react.
(b) Steric factor p: Collision theory gives $A = pZ$, so

$$p = \frac{A}{Z} = \frac{2.0 \times 10^{11}}{4.0 \times 10^{11}} = \boxed{0.50}.$$

Comment: $p = 0.5$ is relatively high (for complex molecules p can be 10^{-1} to 10^{-4}). It indicates a modest orientation requirement, as expected for small, fairly symmetrical reactants. A value below 1 confirms that not every sufficiently energetic collision leads to reaction.
(c) Temperature effect: Raising the temperature broadens the Maxwell–Boltzmann distribution and shifts it to higher energies. Because $f = e^{-E_a/RT}$ depends exponentially on T, even a small rise in T produces a large increase in the area under the curve beyond E_a. Near room temperature, for typical values of E_a ($\approx 50\ \text{kJ mol}^{-1}$), a 10 K rise approximately doubles f, and hence roughly doubles the rate. (The collision frequency Z increases only slightly, as $\sqrt{T}$, and is not the main cause.)
(d) Why $p < 1$: Reacting molecules must collide with the correct orientation for the right bonds to break and form. For example, in $NO + O_3 \longrightarrow NO_2 + O_2$, the nitrogen end of NO

must strike a terminal oxygen of O_3. Collisions with the wrong geometry, or with energy locked in the wrong vibrational modes, do not lead to products even if $E \geq E_a$.

METHOD **Catalysis: Homogeneous and Heterogeneous**

1. Definition: A catalyst increases the rate of a reaction by providing an alternative pathway of lower activation energy. It is not consumed overall, and it does not change ΔH or the position of equilibrium K_c — it only speeds up the attainment of equilibrium.
2. Homogeneous catalysis: catalyst in the same phase as the reactants (usually aqueous or gas).
   - Mechanism: the catalyst reacts to form an intermediate, then is regenerated.
   - Examples: H^+ in ester hydrolysis; Fe^{2+} (or Fe^{3+}) catalysing $S_2O_8^{2-} + 2\,I^- \longrightarrow 2\,SO_4^{2-} + I_2$; NO in the lead-chamber process for $2\,SO_2 + O_2 \longrightarrow 2\,SO_3$.
   - The catalyst concentration appears in the rate equation, e.g. Rate = $k[H^+][\text{ester}]$.
3. Heterogeneous catalysis: catalyst in a different phase (usually a solid catalyst with gaseous or liquid reactants).
   - Steps: adsorption of reactants onto active sites → reaction on the surface → desorption of products.
   - The rate increases with surface area (powder more effective than lumps) and with the number of active sites.
   - Poisoning: impurities (e.g. CO, H_2S, sulfur compounds, lead compounds) adsorb strongly and block active sites, destroying catalytic activity.
   - Examples: Fe in the Haber process; V_2O_5 in the Contact process; Ni or Pt in catalytic hydrogenation; Pt/Rh in catalytic converters.
4. Graphical analysis: on an energy profile diagram, the catalysed route has a lower peak (lower E_a) but the same reactant and product energy levels (same ΔH). On an Arrhenius plot, the catalysed reaction gives a line of smaller slope magnitude.

WORKED EXAMPLE 23 **Catalysis: Homogeneous Kinetics and Heterogeneous Poisoning**

(a) The oxidation of I^- by $S_2O_8^{2-}$ is catalysed by Fe^{2+}: $S_2O_8^{2-} + 2\,I^- \longrightarrow 2\,SO_4^{2-} + I_2$ (slow when uncatalysed). Catalysed mechanism: Step 1: $S_2O_8^{2-} + 2\,Fe^{2+} \longrightarrow 2\,SO_4^{2-} + 2\,Fe^{3+}$ (slow) Step 2: $2\,Fe^{3+} + 2\,I^- \longrightarrow 2\,Fe^{2+} + I_2$ (fast)
(i) Identify the catalyst and the intermediate.
(ii) Write the overall equation and verify that the catalyst is regenerated.

(iii) The rate law is Rate = $k[S_2O_8^{2-}][Fe^{2+}]$. Explain why the rate depends on $[Fe^{2+}]$ but not on $[I^-]$.
(b) In the Haber process, $N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3$ (Fe catalyst), carbon monoxide acts as a poison. (i) Explain the mechanism of poisoning.
(ii) Sketch the energy profile (energy against reaction coordinate) for the catalysed and uncatalysed reactions, labelling E_a(cat), E_a(uncat) and ΔH.

SOLUTION

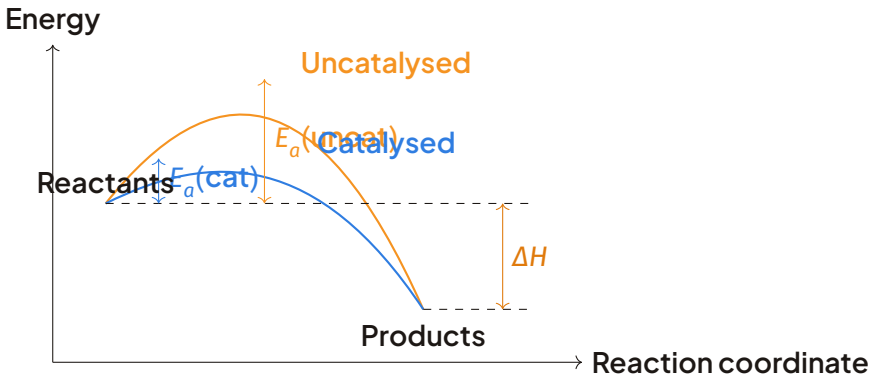
Detailed Solution
(a)(i) Catalyst: Fe^{2+} — it is consumed in Step 1 and regenerated in Step 2. Intermediate: Fe^{3+} — it is formed in Step 1 and consumed in Step 2.
(a)(ii) Overall equation: Add Steps 1 and 2:

$$S_2O_8^{2-} + 2\,Fe^{2+} + 2\,Fe^{3+} + 2\,I^- \longrightarrow 2\,SO_4^{2-} + 2\,Fe^{3+} + 2\,Fe^{2+} + I_2$$

Cancelling Fe^{2+} and Fe^{3+} from both sides:

$$S_2O_8^{2-} + 2\,I^- \longrightarrow 2\,SO_4^{2-} + I_2.$$

The catalyst Fe^{2+} is regenerated. (The uncatalysed reaction is slow because it brings two negative ions, $S_2O_8^{2-}$ and I^-, together; the catalysed route replaces this by steps between ions of opposite charge, which have lower E_a.)
(a)(iii) Rate law interpretation: Step 1 is the slow (rate-determining) step, and it involves $S_2O_8^{2-}$ and Fe^{2+} only. Hence Rate = $k[S_2O_8^{2-}][Fe^{2+}]$. I^- reacts in the fast step after the rate-determining step, so changing $[I^-]$ does not affect the overall rate. Because the catalyst is a reactant in the rate-determining step, its concentration appears in the rate law.
(b)(i) Poisoning mechanism: CO chemisorbs strongly and essentially irreversibly onto the active sites of the iron surface — the very sites where N_2 and H_2 must adsorb and dissociate. The blocked sites are no longer available, the effective surface area for the reaction falls, and the rate drops sharply. Because CO binds more strongly than N_2 or H_2, even trace amounts cause serious loss of activity; the synthesis gas must therefore be carefully purified.
(b)(ii) Energy profile:



Energy profile for an exothermic reaction with and without a catalyst. The catalyst lowers E_a but leaves ΔH unchanged.

Key features: reactants and products are at the same energy levels on both curves, so ΔH is unchanged; the catalysed pathway has a lower transition-state peak, hence a lower E_a and a faster reaction. (For heterogeneous catalysis the profile is often drawn with two smaller humps separated by a shallow minimum corresponding to the adsorbed intermediate.)

METHOD **Physical Methods: Colorimetry, Conductimetry, Gas Syringe and Pressure**

1. Colorimetry / spectrophotometry: based on the Beer–Lambert law, $A = \varepsilon\, b\, c$ (absorbance proportional to concentration).
   - Use when one species is coloured (e.g. I_2, MnO_4^-, $Cr_2O_7^{2-}$, Br_2).
   - Measure absorbance A_t at the wavelength of maximum absorption; $A_t \propto c_t$.
   - Plot A_t against t, or $\ln A_t$ against t to test for first order.
   - Advantages: continuous, rapid, no sampling, very sensitive.
2. Conductimetry: measures the conductivity κ of the mixture, which depends on the concentration and mobility of all ions present.
   - Use for reactions in which the number or nature of ionic species changes, e.g. $CH_3COOC_2H_5 + OH^- \longrightarrow CH_3COO^- + C_2H_5OH$ (conductivity falls because the highly mobile OH^- is replaced by the less mobile CH_3COO^-).
   - Plot κ_t against t; for first order, $\ln(\kappa_t - \kappa_\infty)$ against t is linear.
   - Advantages: continuous, no sampling. Disadvantages: non-specific (all ions contribute) and very sensitive to temperature.
3. Gas syringe / pressure measurement: for reactions involving a gas.
   - Gas evolved (e.g. $CaCO_3 + 2\,HCl$, $2\,H_2O_2 \longrightarrow 2\,H_2O + O_2$): collect the gas in a gas syringe; $V_{gas} \propto n_{gas}$.
   - Change in moles of gas at constant V and T (e.g. $2\,N_2O_5 \longrightarrow 4\,NO_2 + O_2$): $P \propto n_{total}$; use the stoichiometry to relate P_{tot} to the reactant pressure.
4. Dilatometry: follows the small volume change of a liquid reaction mixture in a dilatometer (a flask with a capillary stem); the height of the meniscus is proportional to the extent of reaction. Mentioned for completeness.

WORKED EXAMPLE 24 **Conductimetric Study: Hydrolysis of Ethyl Ethanoate with NaOH**

$CH_3COOC_2H_5(aq) + OH^-(aq) \longrightarrow CH_3COO^-(aq) + C_2H_5OH(aq)$, followed by conductivity measurements. Molar conductivities at the working temperature: $\lambda_{OH^-} = 198$, $\lambda_{CH_3COO^-} = 41$, $\lambda_{Na^+} = 50\ S\ cm^2\ mol^{-1}$; the ester and ethanol are non-ionic. Initial $[OH^-]_0 = [\text{ester}]_0 = 0.010\ mol\ dm^{-3}$; Na^+ is a spectator.

(a) Explain why the conductivity decreases during the reaction.

(b) Derive an expression for the conductivity κ_t at time t in terms of x, the concentration that has reacted, and show that $\kappa_t - \kappa_\infty \propto [OH^-]_t$.

(c) A student claims that, because the reaction is second order, a plot of $\ln(\kappa_t - \kappa_\infty)$ against t should be linear. Discuss this claim and state the correct linear plot.

(d) If the reaction were carried out with OH^- in large excess, what order would be observed? Write the observed rate law and the appropriate linear plot.

SOLUTION

Detailed Solution

(a) Conductivity change: Conductivity depends on the ions present and their mobilities. Na^+ is a spectator and its concentration is constant. During the reaction, highly mobile OH^- ions ($\lambda = 198$) are consumed and replaced by much less mobile CH_3COO^- ions ($\lambda = 41$). The total conductivity therefore decreases as the reaction proceeds.

(b) Expression for κ_t: Let x = concentration reacted ($mol\ dm^{-3}$). Then $[OH^-]_t = 0.010 - x$; $[CH_3COO^-]_t = x$; $[Na^+]_t = 0.010$ (constant).

$$\kappa_t = 10^{-3}\,[(0.010)(50) + (0.010 - x)(198) + (x)(41)] = 10^{-3}\,[2.48 - 157x]\ S\ cm^{-1}.$$

(The factor 10^{-3} converts $mol\ dm^{-3}$ to $mol\ cm^{-3}$.) At $t = 0$ ($x = 0$): $\kappa_0 = 2.48 \times 10^{-3}\ S\ cm^{-1}$. At $t = \infty$ ($x = 0.010$): $\kappa_\infty = 10^{-3}[2.48 - 1.57] = 0.91 \times 10^{-3}\ S\ cm^{-1}$. Then

$$\kappa_t - \kappa_\infty = 10^{-3}\,[(2.48 - 157x) - 0.91] = 157 \times 10^{-3}(0.010 - x) = 157 \times 10^{-3}\,[OH^-]_t.$$

Hence $\kappa_t - \kappa_\infty$ is directly proportional to $[OH^-]_t$.

(c) Discussion of the claim: The claim is incorrect. With equal initial concentrations, $[OH^-] = [\text{ester}]$ throughout, so

$$-\frac{d[OH^-]}{dt} = k[OH^-][\text{ester}] = k[OH^-]^2,$$

a second-order rate law, whose integrated form is

$$\frac{1}{[OH^-]_t} = \frac{1}{[OH^-]_0} + kt.$$

Since $[OH^-]_t \propto \kappa_t - \kappa_\infty$, the correct linear plot is

$$\frac{1}{\kappa_t - \kappa_\infty} \text{ against } t \quad \text{(straight line, slope} \propto k\text{)}.$$

A plot of $\ln(\kappa_t - \kappa_\infty)$ against t is the test for first order and would not be linear here. (Moral: the logarithmic plot is not universal — match the plot to the order being tested.)

(d) Excess OH^- (pseudo-first order): If $[OH^-]_0 \gg [\text{ester}]_0$, then $[OH^-]$ remains effectively constant and

$$\text{Rate} = k[OH^-][\text{ester}] = k_{obs}[\text{ester}], \quad \text{where } k_{obs} = k[OH^-]_0.$$

The observed order is first order (with respect to the ester). In this case the conductivity change now tracks the ester concentration, and a plot of $\ln(\kappa_t - \kappa_\infty)$ against t is linear, with slope = $-k_{obs}$.

10 EXERCISES

10.1 I check my knowledge

EXERCISE 7 **Multiple choice questions**

For each question, choose the single correct answer and justify your choice in one sentence.

1. The unit of the rate of a chemical reaction is:
   (a) mol dm^{-3}
   (b) $\text{mol dm}^{-3}\,\text{s}^{-1}$
   (c) $\text{dm}^{3}\,\text{mol}^{-1}\,\text{s}^{-1}$
   (d) s^{-1}
2. For a reaction which is first order in A and second order in B, the unit of the rate constant k is:
   (a) s^{-1}
   (b) $\text{dm}^{3}\,\text{mol}^{-1}\,\text{s}^{-1}$
   (c) $\text{dm}^{6}\,\text{mol}^{-2}\,\text{s}^{-1}$
   (d) $\text{mol dm}^{-3}\,\text{s}^{-1}$
3. The half-life of a first order reaction:
   (a) depends on the initial concentration of the reactant
   (b) is independent of the initial concentration of the reactant
   (c) doubles when the temperature doubles
   (d) is always equal to the rate constant
4. A catalyst increases the rate of a reaction because it:
   (a) shifts the position of equilibrium towards the products
   (b) increases the average kinetic energy of the particles
   (c) provides an alternative reaction path of lower activation energy
   (d) increases the enthalpy change of the reaction

EXERCISE 8 True or false

State whether each statement is true or false, and justify your answer briefly.

1. The rate of a reaction always increases when the concentration of a reactant increases.
2. The instantaneous rate of a reaction at time t is obtained from the gradient of the tangent to the concentration–time curve at that time.
3. In the titrimetric method of following a reaction, samples withdrawn from the mixture must be quenched before titration.
4. According to collision theory, every collision between reactant particles leads to the formation of products.
5. The Maxwell–Boltzmann distribution shows that raising the temperature increases the proportion of particles with energy greater than the activation energy.

EXERCISE 9 Definitions

Define each of the following terms precisely, as you would in the GCE Advanced Level examination:

1. rate of reaction;
2. order of reaction with respect to a reactant (partial order);
3. overall order of reaction;
4. rate constant;
5. half-life of a reaction;
6. activation energy;
7. activated complex (transition state).

EXERCISE 10 Direct calculations

1. In the decomposition of hydrogen peroxide, $2\,H_2O_2(aq) \longrightarrow 2\,H_2O(l) + O_2(g)$, the concentration of H_2O_2 falls from $0.80\ \text{mol dm}^{-3}$ to $0.50\ \text{mol dm}^{-3}$ in 100 s. Calculate the average rate of disappearance of H_2O_2 over this interval.
2. A first order reaction has a rate constant $k = 2.31e-3\ \text{s}^{-1}$. Calculate its half-life.

3. For the reaction A + B ⟶ products, the rate equation is rate = k[A][B]. Given k = 4.0e - 2 $dm^3\ mol^{-1}\ s^{-1}$, [A] = 0.20 $mol\ dm^{-3}$ and [B] = 0.10 $mol\ dm^{-3}$, calculate the rate of the reaction.

4. The rate constant of a reaction is 1.5e - 4 $dm^6\ mol^{-2}\ s^{-1}$. Deduce the overall order of the reaction.

10.2 I practise

EXERCISE 11 **Average, instantaneous and initial rates**

The decomposition of dinitrogen pentoxide in tetrachloromethane,

$$2\,N_2O_5 \longrightarrow 4\,NO_2 + O_2,$$

was followed by measuring the concentration of N_2O_5 at different times:

t / s	0	200	400	600	800	1000	1200
$[N_2O_5]$ / $mol\ dm^{-3}$	1.00	0.78	0.61	0.47	0.37	0.29	0.22

1. Plot the graph of $[N_2O_5]$ against time.

2. Calculate the average rate of disappearance of N_2O_5 between t = 0 s and t = 400 s, and between t = 400 s and t = 800 s. Comment on the values obtained.

3. By drawing tangents, estimate the instantaneous rate at t = 0 s (initial rate) and at t = 600 s.

4. Show that the ratio of the two instantaneous rates is approximately equal to the ratio of the corresponding concentrations, and state what this suggests about the order of the reaction.

EXERCISE 12 **Initial rate method**

The reaction A + 2 B ⟶ products was studied at 298 K and the following initial rates were measured:

Experiment	$[A]_0$ / $mol\ dm^{-3}$	$[B]_0$ / $mol\ dm^{-3}$	Initial rate / $mol\ dm^{-3}\ s^{-1}$
1	0.10	0.10	2.0×10^{-4}
2	0.20	0.10	4.0×10^{-4}
3	0.10	0.20	1.6×10^{-3}
4	0.20	0.20	3.2×10^{-3}

1. Using the ratio method, determine the order of the reaction with respect to A and with respect to B.
2. Write the rate equation and state the overall order.
3. Calculate the value of the rate constant k, giving its units.
4. Predict the initial rate when $[A]_0 = 0.30\ \text{mol dm}^{-3}$ and $[B]_0 = 0.15\ \text{mol dm}^{-3}$.

WARNING **Check your data before applying the ratio method**

In an initial-rate problem, first verify that the experiments are mutually consistent: the rate predicted for any row from the rate law must match the measured value. If doubling a concentration multiplies the rate by r, the partial order x satisfies $2^x = r$ — and x should normally be a small whole number (0, 1 or 2).

EXERCISE 13 **Arrhenius plot**

The rate constant k of a first order reaction was measured at several temperatures:

T / K	600	650	700	750	800
k / s^{-1}	2.8×10^{-4}	1.6×10^{-3}	7.5×10^{-3}	3.0×10^{-2}	1.0×10^{-1}

1. Starting from the Arrhenius equation $k = A\,e^{-E_a/(RT)}$, show that a graph of $\ln k$ against $\frac{1}{T}$ should be a straight line, and give the expressions of its gradient and its intercept.
2. Complete a table of values of $\frac{1}{T}$ and $\ln k$, then plot the Arrhenius graph.
3. From the gradient, determine the activation energy E_a in kJ mol^{-1} ($R = 8.31\ \text{J K}^{-1}\ \text{mol}^{-1}$).
4. Determine the pre-exponential factor A and comment on the linearity of the graph.

EXERCISE 14 **Conductimetric following of an ester hydrolysis**

The alkaline hydrolysis of an ester,

$$CH_3COOC_2H_5 + OH^- \longrightarrow CH_3COO^- + C_2H_5OH,$$

is followed by measuring the conductivity σ of the mixture as a function of time, the ester and sodium hydroxide being mixed in equal initial concentrations c_0.

1. Explain qualitatively why the conductivity of the mixture decreases as the reaction

proceeds.

2. Sketch the expected shape of the curve $\sigma = f(t)$.

3. The reaction is found to be first order in each reactant. Write the rate equation and show that, since the two initial concentrations are equal, the integrated rate law reduces to

$$\frac{1}{c} = \frac{1}{c_0} + kt.$$

4. Given that σ is proportional to the concentration of the conducting ions, explain how a suitable linear plot allows k to be determined.

5. In one experiment with $c_0 = 0.050\ \text{mol dm}^{-3}$, the plot gave a gradient of $0.12\ \text{dm}^3\ \text{mol}^{-1}\ \text{min}^{-1}$. Deduce the value of k and state its units.

10.3 I prepare for the GCE

EXERCISE 15 **Kinetics of the oxidation of iodide ions (GCE type)**

The reaction between hydrogen peroxide and iodide ions in acidic solution,

$$H_2O_2(aq) + 2\,I^-(aq) + 2\,H^+(aq) \longrightarrow I_2(aq) + 2\,H_2O(l),$$

is studied at 298 K. The iodine liberated is determined by withdrawing $10.0\ \text{cm}^3$ samples at known times, quenching them in ice-cold water, and titrating with standard sodium thiosulfate solution:

$$I_2(aq) + 2\,S_2O_3^{2-}(aq) \longrightarrow 2\,I^-(aq) + S_4O_6^{2-}(aq).$$

1. Explain why the samples are quenched before titration, and state two other methods by which quenching could be achieved. (3 marks)

2. Name the indicator used in the thiosulfate titration and state the colour change at the end point. (2 marks)

3. The following titres of $0.0100\ \text{mol dm}^{-3}$ thiosulfate were obtained:

t / min	0	5	10	15	20	30
$V(S_2O_3^{2-})$ / cm^3	0.0	4.2	7.6	10.4	12.6	15.4

Calculate the concentration of I_2 in the reaction mixture at each time and plot $[I_2]$ against t. (5 marks)

4. From your graph, determine the initial rate of formation of I_2 and the instantaneous rate at $t = 15$ min. (4 marks)

5. Separate initial-rate experiments gave: doubling $[H_2O_2]$ doubles the rate; doubling $[I^-]$ doubles the rate; the rate is independent of $[H^+]$. Deduce the rate equation, the overall order, and the units of k. (4 marks)

6. Suggest a physical method, other than titrimetry, by which this reaction could be followed, and justify your choice. (2 marks)

EXERCISE 16 **First order kinetics and half-life (GCE type)**

The gas-phase decomposition of ethanal,

$$CH_3CHO(g) \longrightarrow CH_4(g) + CO(g),$$

is followed at constant volume by measuring the total pressure in the vessel.

1. Explain why the total pressure can be used to follow the progress of this reaction. (2 marks)

2. The reaction is first order. Starting from the rate equation, derive the integrated rate law

$$\ln \frac{[CH_3CHO]_0}{[CH_3CHO]} = kt,$$

and hence show that the half-life is $t_{1/2} = \dfrac{\ln 2}{k}$. (5 marks)

3. At 800 K, the pressure measurements show that the concentration of ethanal falls to one quarter of its initial value after 420 s. Calculate the rate constant k at this temperature. (3 marks)

4. At 700 K, the rate constant is $4.1e-4\ s^{-1}$. Using the Arrhenius equation, calculate the activation energy of the reaction in kJ mol^{-1} ($R = 8.31\ J\ K^{-1}\ mol^{-1}$). (4 marks)

5. Sketch, on the same axes, the Maxwell–Boltzmann distribution of molecular energies at 700 K and at 800 K, mark the activation energy, and use your sketch to explain why the reaction is faster at the higher temperature. (4 marks)

6. State two reasons, according to collision theory, why most collisions between ethanal molecules do not lead to reaction. (2 marks)

EXERCISE 17 **Energy profiles and catalysis (GCE essay type)**

1. Sketch, on the same diagram, the potential energy profile for an exothermic reaction

occurring (i) without a catalyst and (ii) with a catalyst. Label clearly the activation energy E_a of each path, the enthalpy change of reaction ΔH, and the activated complex. (5 marks)

2. Using transition state theory, describe how the activated complex is formed and explain the role it plays in the conversion of reactants into products. (4 marks)

3. State two similarities and two differences between collision theory and transition state theory. (4 marks)

4. Distinguish between homogeneous and heterogeneous catalysis, giving one named industrial example of each, with a balanced equation for the reaction catalysed. (4 marks)

5. Explain, with reference to the steps of adsorption, reaction and desorption, how a heterogeneous catalyst such as iron in the Haber process lowers the activation energy, and explain why the catalyst can become poisoned. (3 marks)

11 SOLUTIONS TO THE EXERCISES

12 CORRECTED EXERCISES: REACTION KINETICS

SOLUTION **Exercise 1 — Multiple choice questions**

1. Answer (b). A rate is a change of concentration per unit time, so its unit is mol dm^{-3} s^{-1}; option (a) is a concentration, (d) is the unit of a first order rate constant, and (c) is a second order k unit.
2. Answer (c). The overall order is $1 + 2 = 3$; from $k = \frac{\text{rate}}{[A][B]^2}$ the units are $\frac{\text{mol dm}^{-3}\,\text{s}^{-1}}{(\text{mol dm}^{-3})^3} = \text{dm}^6\,\text{mol}^{-2}\,\text{s}^{-1}$.
3. Answer (b). Since $t_{1/2} = \frac{\ln 2}{k}$ contains no concentration term, the half-life of a first order reaction is independent of the initial concentration.
4. Answer (c). A catalyst provides an alternative reaction path of lower activation energy; it changes neither ΔH nor the position of equilibrium, and it does not raise the temperature of the particles.

SOLUTION **Exercise 2 — True or false**

1. False. This is only true when the partial order with respect to that reactant is not zero; for a zero order reactant, the rate is independent of its concentration.
2. True. The instantaneous rate is $\left|\frac{d[X]}{dt}\right|$, which is precisely the gradient of the tangent drawn to the concentration–time curve at time t.
3. True. Quenching (rapid cooling, dilution, or chemical removal of a reactant) stops the reaction in the sample so that the titre reflects the composition at the exact moment of withdrawal.
4. False. Only **effective collisions** — those with energy at least equal to the activation energy and with the correct orientation — lead to products.
5. True. Raising the temperature flattens and shifts the Maxwell–Boltzmann distribution towards higher energies, so the area under the curve beyond E_a (the fraction of particles able to react) increases.

SOLUTION **Exercise 3 — Definitions**

1. Rate of reaction: the change in concentration of a reactant or product per unit time, e.g. rate $= -\frac{d[\mathrm{A}]}{dt}$ for a reactant A; unit $\mathrm{mol\,dm^{-3}\,s^{-1}}$.
2. Partial order: the power to which the concentration of that reactant is raised in the experimentally determined rate equation.
3. Overall order: the sum of the partial orders of all the species appearing in the rate equation.
4. Rate constant: the proportionality constant k in the rate equation; numerically equal to the rate when every reactant concentration is $1\,\mathrm{mol\,dm^{-3}}$, and constant at a given temperature.
5. Half-life: the time taken for the concentration of a reactant to fall to half of its initial value.
6. Activation energy: the minimum energy that colliding particles must possess for a collision to result in reaction.
7. Activated complex (transition state): the unstable, high-energy arrangement of atoms at the maximum of the energy profile, in which old bonds are partially broken and new bonds partially formed; it can decompose either to products or back to reactants.

SOLUTION **Exercise 4 — Direct calculations**

1. Average rate $= -\frac{\Delta[\mathrm{H_2O_2}]}{\Delta t} = \frac{0.80 - 0.50}{100} = 3.0e-3\,\mathrm{mol\,dm^{-3}\,s^{-1}}$.
2. $t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{2.31 \times 10^{-3}} = 300\,\mathrm{s}$.
3. Rate $= k[\mathrm{A}][\mathrm{B}] = 4.0 \times 10^{-2} \times 0.20 \times 0.10 = 8.0e-4\,\mathrm{mol\,dm^{-3}\,s^{-1}}$.
4. For overall order n, the units of k are $\mathrm{dm^{3(n-1)}\,mol^{-(n-1)}\,s^{-1}}$. Here $n - 1 = 2$, so $n = 3$: the reaction is third order overall.

SOLUTION **Exercise 5 — Average, instantaneous and initial rates**

1. The graph of $[\mathrm{N_2O_5}]$ against t is a smooth decay curve, steep at the start and flattening as the concentration falls.

2. Between 0 and 400 s: rate = $\frac{1.00 - 0.61}{400}$ = 9.8*e* − 4 mol dm^{-3} s^{-1}. Between 400 and 800 s: rate = $\frac{0.61 - 0.37}{400}$ = 6.0*e* − 4 mol dm^{-3} s^{-1}. The average rate decreases with time because the concentration of reactant — and hence the frequency of effective collisions — decreases as the reaction proceeds.

3. Tangent at t = 0: gradient ≈ 1.0*e* − 3 mol dm^{-3} s^{-1} (initial rate). Tangent at t = 600 s: gradient ≈ 4.7*e* − 4 mol dm^{-3} s^{-1}. (Values within ±10% are accepted, depending on the tangent drawn.)

4. $\frac{\text{rate}_0}{\text{rate}_{600}} \approx \frac{1.0 \times 10^{-3}}{4.7 \times 10^{-4}} \approx 2.1$ and $\frac{[N_2O_5]_0}{[N_2O_5]_{600}} = \frac{1.00}{0.47} \approx 2.1$. Since rate $\propto [N_2O_5]$, the reaction is first order in N_2O_5.

SOLUTION **Exercise 6 — Initial rate method**

Note on the data. Comparing experiments 1 and 3 ([A] constant, [B] doubled): the rate is multiplied by $\frac{1.6 \times 10^{-3}}{2.0 \times 10^{-4}} = 8 = 2^3$, which would give order 3 in B; but experiments 2 and 4 then give $\frac{3.2 \times 10^{-3}}{4.0 \times 10^{-4}} = 8$ for doubling both concentrations, which is inconsistent ($2^1 \times 2^3 = 16 \neq 8$). The table as printed is therefore internally inconsistent. The intended, consistent data are: experiment 3, rate = 8.0×10^{-4}; experiment 4, rate = 1.6×10^{-3} mol dm^{-3} s^{-1}. These corrected values are used below.

1. Experiments 1 and 2 ([B] constant): doubling [A] doubles the rate, so $2^x = 2$, giving order 1 in A. Experiments 1 and 3 ([A] constant): doubling [B] multiplies the rate by $\frac{8.0 \times 10^{-4}}{2.0 \times 10^{-4}} = 4 = 2^2$, so order 2 in B. Check with experiment 4: $2^1 \times 2^2 = 8$ and $\frac{1.6 \times 10^{-3}}{2.0 \times 10^{-4}} = 8$ — consistent.

2. Rate equation: rate = k[A][B]^2; overall order = 1 + 2 = 3.

3. $k = \frac{\text{rate}}{[A][B]^2} = \frac{2.0 \times 10^{-4}}{0.10 \times (0.10)^2} = 0.20$ dm^6 mol^{-2} s^{-1} (third order units).

4. Rate = $0.20 \times 0.30 \times (0.15)^2$ = 1.35*e* − 3 mol dm^{-3} s^{-1}.

Examiner's advice: always verify that every row of the table satisfies the rate law you propose; if doubling a concentration multiplies the rate by r, the partial order x obeys $2^x = r$, and x should be a small whole number (0, 1 or 2).

SOLUTION **Exercise 7 — Arrhenius plot**

1. Taking natural logarithms of $k = A\,e^{-E_a/(RT)}$:

$$\ln k = \ln A - \frac{E_a}{R}\cdot\frac{1}{T}.$$

This is of the form $y = mx + c$ with $y = \ln k$ and $x = \frac{1}{T}$: a straight line of gradient $-\frac{E_a}{R}$ and intercept $\ln A$.

2.

$\frac{1}{T}$ / K^{-1} ($\times 10^3$)	1.667	1.538	1.429	1.333	1.250
$\ln k$	-8.18	-6.44	-4.89	-3.51	-2.30

The five points lie on a good straight line of negative gradient.

3. Gradient $= \dfrac{-2.30 - (-8.18)}{(1.250 - 1.667)\times 10^{-3}} = \dfrac{5.88}{-4.17\times 10^{-4}} \approx -1.41e4$ K. Hence

$$E_a = 1.41\times 10^4 \times 8.31 \approx 1.17e5\ \text{J mol}^{-1} = 117\ \text{kJ mol}^{-1}.$$

4. Intercept: $\ln A \approx -8.18 + 1.41\times 10^4 \times 1.667\times 10^{-3} \approx 15.3$, so $A \approx e^{15.3} \approx 4.4e6\ \text{s}^{-1}$. The excellent linearity confirms that the reaction obeys the Arrhenius equation over this temperature range, i.e. E_a is effectively constant.

SOLUTION **Exercise 8 — Conductimetric following of an ester hydrolysis**

1. The highly mobile OH^- ions (very high molar ionic conductivity) are consumed and replaced by CH_3COO^- ions, which conduct much less; the ester and ethanol are non-ionic. The total ionic conductance therefore falls as the reaction proceeds.

2. σ starts high, decreases rapidly at first, then more and more slowly, and levels off at a non-zero final value σ_∞ due to the CH_3COO^- and Na^+ ions remaining in solution.

3. Rate $= k[\text{ester}][OH^-]$. Since the initial concentrations are equal and the reactants are consumed in a 1 : 1 ratio, $[\text{ester}] = [OH^-] = c$ at all times, so

$$-\frac{dc}{dt} = kc^2 \quad\Longrightarrow\quad -\frac{dc}{c^2} = k\,dt.$$

Integrating between $(c_0, 0)$ and (c, t): $\frac{1}{c} - \frac{1}{c_0} = kt$, i.e.

$$\frac{1}{c} = \frac{1}{c_0} + kt,$$

the integrated second order rate law.

4. Since $\sigma \propto c$ (the concentration of conducting ions), $\frac{1}{\sigma} \propto \frac{1}{c}$. A plot of $\frac{1}{\sigma}$ against t is

therefore a straight line whose gradient is proportional to k; calibration with solutions of known concentration converts the gradient into k.

5. The gradient of the $\frac{1}{c}$ versus t plot is k, so $k = 0.12\ \text{dm}^3\ \text{mol}^{-1}\ \text{min}^{-1}$ (second order units), equivalently $2.0e-3\ \text{dm}^3\ \text{mol}^{-1}\ \text{s}^{-1}$.

SOLUTION **Exercise 9 — Kinetics of the oxidation of iodide ions (GCE type)**

1. Quenching stops (or slows to a negligible rate) the reaction in the withdrawn sample, so the titre measures the I_2 present at the exact time of sampling. (1) Other quenching methods: dilution with a large volume of ice-cold water (1); chemical quenching, e.g. neutralising the acid or adding a reagent that rapidly consumes one reactant (1).

2. Indicator: freshly prepared starch solution, added near the end point (1); colour change from blue-black to colourless (1).

3. $n(S_2O_3^{2-}) = 0.0100 \times \frac{V}{1000}$; from the stoichiometry, $n(I_2) = \frac{1}{2}n(S_2O_3^{2-})$ in the 10.0 cm^3 sample, so $[I_2] = \frac{n(I_2)}{0.0100\ \text{dm}^3}$. (2 for method)

t / min	0	5	10	15	20	30
$[I_2]$ / mol dm^{-3} ($\times 10^3$)	0.0	2.1	3.8	5.2	6.3	7.7

(2 for the values) The graph is a curve rising steeply at first and flattening as the reactants are consumed. (1 for the plot)

4. Initial rate (tangent at the origin) $\approx \frac{2.1 \times 10^{-3}}{5} \approx 4.2e-4\ \text{mol}\ \text{dm}^{-3}\ \text{min}^{-1}$ (2); tangent at $t = 15$ min gives $\approx 2.4e-4\ \text{mol}\ \text{dm}^{-3}\ \text{min}^{-1}$ (2). (Values consistent with the candidate's own tangents are accepted.)

5. First order in H_2O_2, first order in I^-, zero order in H^+ (2): rate $= k[H_2O_2][I^-]$; overall order = 2 (1); units of k: $\text{dm}^3\ \text{mol}^{-1}\ \text{s}^{-1}$ (1).

6. Colorimetry (1): iodine is coloured (yellow-brown, or blue-black with starch), so the absorbance at a suitable wavelength is proportional to $[I_2]$ and can be recorded continuously without sampling (1).

Examiner's expectations: correct use of the 1 : 2 stoichiometry between I_2 and $S_2O_3^{2-}$; axes labelled with quantities and units; tangents drawn with a ruler; units quoted for every rate.

SOLUTION **Exercise 10 — First order kinetics and half-life (GCE type)**

1. One mole of gaseous reactant gives two moles of gaseous products, so the number of moles of gas — and hence the total pressure at constant volume and temperature — increases as the reaction proceeds; the pressure rise is proportional to the extent of reaction. (2)

2. Rate $= -\dfrac{d[CH_3CHO]}{dt} = k[CH_3CHO]$ (1). Separating variables: $-\dfrac{d[CH_3CHO]}{[CH_3CHO]} = k\,dt$ (1). Integrating from $[CH_3CHO]_0$ at $t = 0$ to $[CH_3CHO]$ at time t:

$$\ln[CH_3CHO]_0 - \ln[CH_3CHO] = kt \quad \Longrightarrow \quad \ln\frac{[CH_3CHO]_0}{[CH_3CHO]} = kt \quad (1).$$

At $t = t_{1/2}$, $[CH_3CHO] = \dfrac{[CH_3CHO]_0}{2}$ (1), so $\ln 2 = k\,t_{1/2}$ and $t_{1/2} = \dfrac{\ln 2}{k}$ (1).

3. Falling to one quarter means two half-lives: $t_{1/2} = 210$ s (1), so $k = \dfrac{0.693}{210} = 3.3e-3\ s^{-1}$ (2).

4. $\ln\dfrac{k_{800}}{k_{700}} = \dfrac{E_a}{R}\left(\dfrac{1}{700} - \dfrac{1}{800}\right)$ (1). $\ln\dfrac{3.3\times10^{-3}}{4.1\times10^{-4}} = \ln 8.05 = 2.09$ (1); $\dfrac{1}{700} - \dfrac{1}{800} = 1.79\times10^{-4}\ K^{-1}$ (1). Hence

$$E_a = \frac{2.09 \times 8.31}{1.79 \times 10^{-4}} \approx 9.7e4\ J\,mol^{-1} = 97\ kJ\,mol^{-1} \quad (1).$$

5. Both curves start at the origin, rise to a maximum and fall asymptotically; the 800 K curve is lower, flatter and shifted towards higher energies (2). With E_a marked on the energy axis, the area under the curve beyond E_a — the fraction of molecules energetic enough to react on collision — is markedly larger at 800 K, so the frequency of effective collisions, and hence the rate, is greater (2).

6. (i) Most colliding molecules have kinetic energy less than the activation energy (1); (ii) many collisions occur with an unsuitable orientation for bond breaking and bond making (1).

Examiner's expectations: a clear derivation with correct limits of integration; the Arrhenius ratio form used correctly with temperatures in kelvin; Maxwell–Boltzmann curves that both start at the origin and never touch the axis again.

SOLUTION **Exercise 11 — Energy profiles and catalysis (GCE essay type)**

1. The diagram shows potential energy (vertical axis) against reaction coordinate: reactants at one level, products at a lower level ($\Delta H < 0$) (1). The uncatalysed path rises to a single high peak (the activated complex) with activation energy E_a (2); the catalysed

path passes through one or more lower peaks with activation energy $E_a' < E_a$ (1); ΔH is identical for both paths (1).

2. As reactant particles approach with sufficient energy, old bonds begin to break while new bonds begin to form; at the top of the energy barrier the atoms adopt a partially bonded, unstable configuration called the activated complex (transition state) (2). It may either fall apart back into reactants or proceed forward to products; the rate of reaction is governed by the concentration of activated complexes and the frequency with which they cross the barrier (2).

3. Similarities (2): both require a minimum energy (the activation energy) for reaction; both explain the increase of rate with temperature through the greater number of species able to cross the energy barrier. Differences (2): collision theory treats reaction as occurring at the instant of an effective collision and uses the Maxwell–Boltzmann distribution, whereas transition state theory postulates a definite, short-lived intermediate species (the activated complex) at the energy maximum; transition state theory accounts for orientation through the structure of the complex (entropy of activation), while collision theory introduces orientation only through a steric factor.

4. Homogeneous catalysis: catalyst and reactants in the same phase (1) — e.g. the acid-catalysed hydrolysis of an ester, $CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$ (1). Heterogeneous catalysis: catalyst in a different phase from the reactants (1) — e.g. the Haber process, $N_2(g) + 3\,H_2(g) \xrightarrow{Fe} 2\,NH_3(g)$, solid iron catalysing gaseous reactants (1). (Also accepted: V_2O_5 in the Contact process, $2\,SO_2(g) + O_2(g) \longrightarrow 2\,SO_3(g)$; Ni in the hydrogenation of alkenes.)

5. Adsorption: N_2 and H_2 molecules are adsorbed onto the iron surface, which weakens the strong N ≡ N and H – H bonds; reaction: the adsorbed atoms are held close together in favourable orientations and combine stepwise to NH_3 along a path of lower activation energy; desorption: the ammonia molecules leave the surface, freeing the active sites (2). Poisoning: impurities such as sulfur compounds or carbon monoxide adsorb strongly and irreversibly on the active sites, blocking them and destroying the catalytic activity (1).

Examiner's expectations: a clearly labelled energy profile with both paths and an unchanged ΔH; precise vocabulary (adsorption, active sites, desorption, poison); balanced equations with the catalyst written over the arrow.

13 SUMMARY SHEET



Mind map of the chapter Reaction Kinetics.

KEY POINTS **Reaction Kinetics — Summary Sheet**

1. Key definitions

- **Rate of reaction** v: for a reaction $\nu_A A + \nu_B B \rightarrow \nu_C C + \nu_D D$, $v = -\frac{1}{\nu_A}\frac{d[A]}{dt} = -\frac{1}{\nu_B}\frac{d[B]}{dt} = +\frac{1}{\nu_C}\frac{d[C]}{dt} = +\frac{1}{\nu_D}\frac{d[D]}{dt}$. Usual unit: $\text{mol dm}^{-3}\,\text{s}^{-1}$.
- **Average rate**: gradient of the chord between two points of a concentration–time curve; **instantaneous rate**: gradient of the tangent at time t; **initial rate**: instantaneous rate at $t = 0$.
- **Rate law**: experimentally determined equation $v = k[A]^m[B]^n$.
- **Rate constant** k: proportionality constant, dependent only on temperature (and catalyst).
- **Order** with respect to A = m; **overall order** = $m + n$. Orders are experimental, not stoichiometric coefficients; they can be fractional or negative.
- **Half-life** $t_{1/2}$: time for the concentration of a reactant to fall to half its initial value.
- **Activation energy** E_a: minimum energy colliding particles must possess to react.
- **Catalyst**: substance that increases the rate by providing a path of lower E_a, without being consumed.

2. Laws and formulas to know

- Rate law: $v = k[A]^m[B]^n$; units of k: $\text{mol}^{1-(m+n)}\,\text{dm}^{3(m+n)-3}\,\text{s}^{-1}$.
- First order: $[A]_t = [A]_0 e^{-kt}$; $\ln[A]_t = \ln[A]_0 - kt$; $t_{1/2} = \frac{\ln 2}{k} \approx \frac{0.693}{k}$ (independent of $[A]_0$).
- Zero order: $[A]_t = [A]_0 - kt$ (straight line of [A] vs t); $t_{1/2} = \frac{[A]_0}{2k}$.
- Second order (single reactant): $\frac{1}{[A]_t} = \frac{1}{[A]_0} + kt$; $t_{1/2} = \frac{1}{k[A]_0}$.
- Arrhenius equation: $k = A\,e^{-E_a/(RT)}$, i.e. $\ln k = \ln A - \frac{E_a}{RT}$; plot of $\ln k$ vs $\frac{1}{T}$ gives slope $-\frac{E_a}{R}$.
- Two-temperature form: $\ln\frac{k_2}{k_1} = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$.

3. Methods of measuring rates

- **Chemical (titrimetric)**: withdraw samples at known times, quench (cool, dilute or remove catalyst), then titrate. Example: H_2O_2 decomposition followed by titration with $KMnO_4$.
- **Colorimetry**: absorbance proportional to concentration (Beer–Lambert law $A = \varepsilon c l$); suited to coloured species such as I_2 or MnO_4^-.
- **Conductimetry**: follows changes in ionic concentration (e.g. ester hydrolysis producing H^+); **dilatometry**: volume change in a dilatometer.

- **Gas syringe / pressure**: volume of gas evolved vs time (e.g. $CaCO_3$ + HCl); for gases at constant volume, pressure is proportional to amount of gas.
- **Initial-rate method**: measure v_0 for several experiments varying one concentration at a time; compare ratios to find each partial order.
- **Graphical method**: test linearity of [A] vs t (order 0), ln[A] vs t (order 1), $\frac{1}{[A]}$ vs t (order 2); constant half-life ⇒ first order.

4. Theories

- **Collision theory**: reaction requires collisions with energy ≥ E_a and correct orientation; rate increases with concentration, temperature (Maxwell–Boltzmann distribution shifts), surface area.
- **Transition-state theory**: reactants pass through a high-energy **activated complex** at the top of the energy profile; $\Delta H = E_a$(forward) − E_a(reverse) for elementary steps.

5. Catalysis

- **Homogeneous**: catalyst and reactants in the same phase (e.g. Fe^{2+} catalysing $S_2O_8^{2-}/I^-$).
- **Heterogeneous**: different phases; reactants adsorb onto the surface, react, then desorb (e.g. Fe in the Haber process, Ni in hydrogenation).
- A catalyst lowers E_a, is unchanged at the end, and does not shift the position of equilibrium.

6. Common mistakes

- Confusing order with stoichiometric coefficients — orders come only from experiment.
- Forgetting that units of k depend on the overall order.
- Using $t_{1/2} = \ln 2 / k$ for reactions that are not first order.
- Forgetting to quench samples in titrimetric methods (reaction continues!).
- Saying a catalyst "increases the yield" — it only increases the rate.
- Mixing units: convert cm^3 to dm^3, minutes to seconds when required.

7. GCE tips

- Always draw the tangent carefully and show the gradient triangle when reading rates from curves.
- In initial-rate problems, tabulate experiments and divide rates where one concentration is constant.
- State units for every quantity; give k with its correct units.
- For Arrhenius plots, quote the slope = $-E_a/R$ and compute E_a in kJ mol^{-1}.

Signed • OnBuch+
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